Chiziqli algebra I

1–4 haftalar

Har bir kurs fayli va tayinlangan darslik bo‘limlaridagi barcha savollar va formulalar — vektorlar, chiziqli kombinatsiyalar va span; skalyar va vektor ko‘paytmalar; matritsalar, matritsa algebrasi, determinantlar; rang va teskari matritsa — har bir masala to‘liq bosqichma-bosqich yechim bilan. Mustaqil o‘rganish va Test 1 ga tayyorgarlik uchun tuzilgan Toshkent texnologiya universiteti $\cdot 2026$-yil Kuz Manbalar: kurs konspektlari, darslar, ISA, TA sessiyalari; G. Strang, Introduction to Linear Algebra $(4e)$ va Linear Algebra and its Applications $(4e) SAVOLLAR BAZASI \cdot FORMULALAR KATALOGI \cdot TO‘LIQ YECHIMLAR OKTABR 2026$

Manbalar: kurs konspektlari, darslar, ISA, TA sessiyalari; G. Strang, Introduction to Linear Algebra $(4e)$ va Linear Algebra and its Applications $(4e)$

Contents

1 Kurs xaritasi va hujjatdan qanday foydalanish 2

2 Hafta 1 — Vektorlar (Chiziqli algebra I) 2

2.1 Chiziqli algebra I: 1-hafta formulalari katalogi 2

2.2 Chiziqli algebra I, 1-hafta: ISA mashqlari — savollar va to‘liqyechimlar . 2

2.3 Chiziqli algebra I, hafta 1: TA sessiya masalalari (alternativto’plam) — savollar va to’liq yechimlar 8

3 2-Hafta — Skalyar va vektor ko’paytmalari,matritsalarga kirish (Lineal algebra I) 10

3.1 Lineal algebra I, 2-hafta: formulalar katalogi 10

3.2 Lineal algebra I, 2-hafta: ma’ruzadagi auditoriya masalalari —savollar va to’liq yechimlar 10

3.3 Chiziqli algebra I, nedelya 2: mashqlar ISA — savollar va to‘liqyechimlar 13

3.4 Chiziqli algebra I, 2-hafta: TA sessiyasi (15 masala) — savollarva to’liq yechimlar 18

4 3-hafta — Matritsa algebrasi, determinantlar, aralashko’paytma (Chiziqli algebra I) 21

4.1 Chiziqli algebra I: 3-hafta formulalari katalogi 21

4.2 Chiziqli algebra I: 3-hafta ma’ruzasi masalalari (15) — savollarva to’liq yechimlar 22

5 4-hafta — Matritsa rangi va teskari matritsa (Linealalgebra I) 28

5.1 4-hafta formulalar katalogi 29

5.2 4-hafta mashqlari (rang, teskari, Gauss-Jordan) — savollar vato’liq yechimlar 29

6 Darsliklardan qo’shimchalar — to’liq yechimlari bilanmajburiy o’qish uchun qo’shimcha mashqlar 32

6.1 Strang, Introduction to Linear Algebra 4e — tanlanganmasalalar yechilgan 32

6.2 Strang, Linear Algebra and its Applications 4e, §1.6 (Teskari va-transponirlangan) — asosiy natijalar va tahlil qilingan misol 33

1 Kurs xaritasi va hujjatdan qanday foydalanish

Hujjat bu 2026-yil kuz semestrining birinchi choragi bo‘lgan ”Chiziqli algebra I” kursining birinchi to‘rt haftasiga mo‘ljallangan to‘liq va mustaqbil qo‘llanmadir. U kurs repositoriyasining course-files/ papkasidagi har bir tegishli manbadan qo‘lda tuzilgan: haftalik konpektlar (Notes), ma’ruzalar prezentatsiyalari, amaliy mashg‘ulotlar varaqalari, mustaq ish uch vazifalar $(ISA), TA$ sessiyalarining masalalari va G. Strang darsliklarining belgilangan bo‘limlari. Masalalar so‘zma-so‘z takrorlangan; yechimlar to‘liq, bosqichma-bosqich, tekshirishlar bilan.

Hafta	Chiziqli algebra I
1	Vektorlar; chiziqli kombinatsiyalar; span; chiziqli mustaqillik
2	Skalyar ko‘paytma, burchaklar, proyeksiyalar; vektor ko‘paytma, yuzalar; matritsalar: amallar
3	Matritsa algebrasi; transponir; matritsalarni ko‘paytirish; determinantlar; aralash ko‘paytma
4	Matritsa rangi; Gauss-Jordan usulida teskari matritsa

Yozuv. Matritsat satr bo‘yicha yoziladi: $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ — birinchi satr ab, ikkinchi cd; ustun-vektor $\begin{bmatrix} x \\ y \end{bmatrix}$, satr-vektor $\begin{bmatrix} x & y \end{bmatrix}$ ko‘rinishida yoziladi. Darajalar va indekslar ^{} va_ orqali yoziladi: A^{-1}, A^T, a_{11}.

2 Hafta 1 — Vektorlar (Chiziqli algebra I)

2.1 Chiziqli algebra I: 1-hafta formulalari katalogi

Vektorlar R^n dagi ustunlardir; butun 1-hafta qo‘shish, skalyar ko‘paytirish, kombinatsiyalar, span va mustaqillikdan tarkib topgan. $(u+v)_i = u_i + v_i, (cu)_i = cu_i$(komponent bo‘yicha amallar)$(24) c_1v_1 + \cdots + c_nv_n =$ chiziqli kombinatsiya, (25)span$\{v_1, \ldots, v_n\} =$ bunday barcha kombinatsiyalar to‘plami, $(26) v_1, \ldots, v_n$ mustaqil $\Longleftrightarrow c_1v_1 + \cdots + c_nv_n = 0 \Longrightarrow c_1 = \cdots = c_n = 0$, $(27) qx \|x\| = x_1^2 + \cdots + x_n^2$, $x\hat{} = \|x\|$(birlashuvchi vektor)$(28) v = Luv$ vektori L uzunlikda u bo‘ylab : $\|u\|, (29)$

$$d(P, Q) = \|Q - P\|. (30)$$

2.2 Chiziqli algebra I, 1-hafta: ISA mashqlari — savollar va to‘liq yechimlar

$$[][] - 2 - 1$$

Butun vektorlar bo‘limida vektorlar ustunlardir; A qismida $Au =, v =$. 3 5

$$[][] - 2 - 1$$

1. $u = \begin{bmatrix} -2 \\ 3 \end{bmatrix}, v = \begin{bmatrix} -1 \\ 5 \end{bmatrix}$ **bo‘lsin.** $u + v, u - v$**, 2u, −3v, 2u − 3v ni hisoblang.**

[Manba: $ISALA$, hafta 1, mashq 1]

Yechim. (24) bo‘yicha komponent bo‘yicha: $u + v = \begin{bmatrix} -3 \\ 8 \end{bmatrix}$; $u - v = \begin{bmatrix} -1 \\ -2 \end{bmatrix}$; 2u $= \begin{bmatrix} -4 \\ 6 \end{bmatrix}$; $-$3v $= \begin{bmatrix} 3 \\ -15 \end{bmatrix}$; 2u $-$ 3v $= \begin{bmatrix} -4 & + & 3 \\ 6 & + & 15 \end{bmatrix} = \begin{bmatrix} -1 \\ 21 \end{bmatrix}$.

2. Berilgan $a = \begin{bmatrix} -1 \\ 4 \end{bmatrix}, b = \begin{bmatrix} 2 \\ -2 \end{bmatrix}$**; toping** $a + b, a - b$**, 3a + 2b.**

$$\begin{bmatrix} 5 & -1 \\ 6 & 12 \end{bmatrix}$$

[Manba: $ISALA$, hafta 1, mashq 2]

Yechim. $a+b = \begin{bmatrix} -1 & + & 2 \\ 4 & & 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$**;** $a-b = \begin{bmatrix} -1 & & 2 \\ 4 & + & 2 \end{bmatrix} = \begin{bmatrix} -3 \\ 6 \end{bmatrix}$**; 3a + 2b =** $\mathbf{[}3\cdot(-1)+2\cdot 2$**;** $3 \cdot 4 + 2 \cdot (-2)\mathbf{]} = \begin{bmatrix} 1 \\ 8 \end{bmatrix}$**.**

$$\begin{bmatrix} 2 & 6 & 12 \\ -4 & 8 & \end{bmatrix} \begin{bmatrix} 2 \\ 7 \end{bmatrix}$$

3. Toping x**, agar** $x + \begin{bmatrix} 2 \\ -4 \\ 1 \end{bmatrix} = \begin{bmatrix} 7 \\ 3 \\ -2 \end{bmatrix}$**.**

[Manba: $ISALA$, hafta 1, mashq 3]

Yechim. Ma**’lum vektorni ikkala tomondan ayiramiz:** $x = \begin{bmatrix} 7 & & 2 \\ 3 & + & 4 \\ -2 & & 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 7 \\ -3 \end{bmatrix}$**.**

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} - \begin{bmatrix} 2 \\ -4 \\ 1 \end{bmatrix} = \begin{bmatrix} 8 \\ 3 \\ -2 \end{bmatrix}$$

4. Toping x**, agar 3x − 2u** $= v$**, bunda** $u = \begin{bmatrix} -2 \\ 1 \end{bmatrix}, v = \begin{bmatrix} 8 \\ 2 \end{bmatrix}$**.**

[Manba: $ISALA$, hafta 1, mashq 4]

Yechim. Ajratamiz: 3x $= v+$ 2u $= \begin{bmatrix} 8 & & 4 \\ 2 & + & 2 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \end{bmatrix}$, shundan $x =$ [4/3; 4/3]. Skalyarga ko’paytirish qo’shishga nisbatan taqsimlanuvchi, shuning uchun vektorni $3ga$ bo’lish degani, uning har bir komponentini $3ga$ bo’lishdir.

5. Vektorlarning Evklid uzunligini hisoblang: $u = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$, $v = \begin{bmatrix} 2 \\ -3 \\ 6 \end{bmatrix}$, $w = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$.

[Manba: $ISALA$, hafta 1, mashq 5]

Yechim. $(28)ga$ ko'ra: $\|u\| = \sqrt{9} + 16 = 5$; $\|v\| = \sqrt{4} + 9 + 36 = 7$; $\|w\| = 0$(faqat nol vektor nol uzunlikka ega).

6. $v = \begin{bmatrix} -4 \\ 3 \end{bmatrix}$ **vektoriga bir xil yo'nalishda birlik vektorni toping va uning uzunligi 1 ga teng ekanini tekshiring.**

$$[-\frac{4}{5}; \frac{3}{5}]$$

[Manba: $ISALA$, hafta 1, mashq 6]

Yechim. $\|v\| = \sqrt{16} + 9 = 5$**, shuning uchun** $\hat{v} = v/5 =$ **[−4/5; 3/5]. Tekshirish:** $\|\hat{v}\| = \sqrt{\frac{16}{25} + \frac{9}{25}} = 1$**.**

7. $v = \begin{bmatrix} 3 \\ -8 \end{bmatrix}$ **vektoriga bir xil yo'nalishda uzunligi 10 bo'lgan vektorni toping.**

[Manba: $ISALA$, hafta 1, mashq 7]

Yechim. Bu yerda $\|v\| = 5$, shuning uchun $\frac{10}{5} = 2$ miqdorida masshtablaymiz: javob 2v =

$$\begin{bmatrix} 6 \\ -16 \end{bmatrix}$$

$\|2v\| = 2 \cdot 5 = 10$ norma bir xilligi bo'yicha.

8. Vektorlarni ayirish va uzunlikdan foydalanib, $P = (1, -2, 3) va Q = (5, 1, -3)$ **orasidagi masofani toping.**

[Manba: $ISALA$, hafta 1, mashq 8]

Yechim. $PQ = Q - P = \begin{bmatrix} 5 & & 1 \\ 1 & + & 2 \\ -3 & & 3 \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \\ -6 \end{bmatrix}$**, shuning uchun** $d(P, Q) = \|PQ\| = \sqrt{16} + 9 + 36 = \sqrt{61}$**.**

$$[5 - 1; 1 - (-2); -3 - 3]$$

$$\begin{bmatrix} 4 \\ 3 \\ -6 \end{bmatrix}$$

. Shuning uchun $d(P, Q) = \|PQ\| = \sqrt{4^2} + 3^2 + (-6)^2 = \sqrt{16} + 9 + 36 = \sqrt{61}$.

9. $v_1 =, v_2 =$ **bo'lsin.** $2v_1 + v_2, -v_1 + 3v_2, 4v_1 - 2v_2, 0v_1 + 5v_2$ **ni hisoblang. 2 3**

$$[][][][][]1 - 1 - 1 - 24 + 26$$

[Manba: $ISALA$, hafta 1, mashq 9]

Yechim. $2v_1 + v_2 =$

$$\begin{bmatrix} 2 & 2 \\ 6 & 3 \end{bmatrix}$$

; $-v_1 + 3v_2 =$

$$\begin{bmatrix} -1 & 3 \\ -3 & 9 \end{bmatrix}$$

; $4v_1 - 2v_2 =$

$$\begin{bmatrix} 4 & -2 \\ 12 & -6 \end{bmatrix}$$

; $0v_1 + 5v_2 = (nol$ skalyar v_1ni to'liq yo'q qiladi).

$$\begin{bmatrix} 0 & 5 \\ 0 & 15 \end{bmatrix}$$

$5v_2 =$

$$\begin{bmatrix} 1 & -1 & 7 \\ -1 & -2 & 4 \end{bmatrix} + \begin{bmatrix} 2 & 6 \\ 0 & 0 \end{bmatrix}$$

.

10. c_1, c_2 **ni toping shundayki** $c_1 + c_2 =$**; almashtirish orqali tekshiring. 1 2 1**

[Manba: $ISALA$, hafta 1, mashq 10]

Yechim. Komponentli tenglamalar: $c_1 - c_2 = 7 va c_1 + 2c_2 = 1$. Birinchisini ikkinchisidan ayiramiz:

$$c_1 + c_2 = 1; c_1 - c_2 = 7; 2c_1 + c_2 = 1.$$

$3c_2 = -6$, demak $c_2 = -2 va c_1 = 5$. Tekshirish:

$$\begin{bmatrix} 5 & & 2 \\ 5 & + & 2 \end{bmatrix} = \begin{bmatrix} 7 \\ 1 \end{bmatrix}$$

.

11. $b = \begin{bmatrix} 5 \\ 1 \\ 4 \end{bmatrix}$ **vektorini** $b = c1 \cdot v1 + c2 \cdot v2 + c3 \cdot v3$ **ko'rinishida yozish mumkinmi, bunda** $v1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$, $v2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$, $v3 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$**?**

[Manba: $ISALA$, hafta 1, mashq 11]

Yechim. Komponentlar $c_1 + c_3 = 5, c_2 + c_3 = 1, c_1 + c_2 = 4$ beradi. Birinchi ikkitasini qo'shib, uchinchisini ayiramiz: $2c_3 = 2$, demak $c_3 = 1, c_1 = 4, c_2 = 0$. Kombinatsiya

mavjud:$b = 4v_1 + 0 \cdot v_2 + 1 \cdot v_3$. Yechimning noyobligi shuni ham ko'rsatadiki, uchta vektor chiziqli bog'liq emas.

12. $c \cdot \begin{bmatrix} -2 \\ 3 \end{bmatrix}, c \in R$ **bo'lgan barcha vektorlarni geometrik tasvirlang;** $\begin{bmatrix} 4 \\ -6 \end{bmatrix}$, $\begin{bmatrix} -2 \\ 3 \end{bmatrix}$, $\begin{bmatrix} 6 \\ -8 \end{bmatrix}$ **vektorlari bu** span **ga tegishlimi?**

[Manba: $ISALA$, hafta 1, mashq 12]

Yechim. Span — boshlang'ich nuqtadan o'tuvchi (−2, 3) yo'nalishidagi butun to'g'ri chiziq. Tekshirish: $\begin{bmatrix} 4 \\ -6 \end{bmatrix}$ — ha $(c = -2 : -2 \cdot (-2) = 4va - 2 \cdot 3 = -6)$; $\begin{bmatrix} -2 \\ 3 \end{bmatrix}$ — ha $(c = 1)$; $\begin{bmatrix} 6 \\ -8 \end{bmatrix}$ uchun −2c = 6 va 3c = −8, ya'ni $c = -3vac = -\frac{8}{3}$ — ziddiyat, demak u to'g'ri chiziqda yotmaydi.

13. $v_1 =, v_2 =$ **vektorlari chiziqli mustaqilmi?** $c_1v_1 + 26c_2v_2 = 0$ **bo'lgan nol bo'lmagan** c_1, c_2 **ni toping va geometrik tushuntiring.**

[Manba: $ISALA$, hafta 1, mashq 13]

Yechim. $v_2 = 3v_1$, shuning uchun $3v_1 - v_2 = 0$ — no trivial bog'liqlik $(c_1 = 3, c_2 = -1)$. Vektorlar bog'liq. Geometrik jihatdan ikkalasi ham boshlang'ich nuqtadan o'tuvchi bir to'g'ri chiziqda yotadi, shuning uchun ular birga hali ham faqat shu to'g'ri chiziqni hosil qiladi — bog'liqlik bitta vektor yangi yo'nalish bermasligini anglatadi.

$$\begin{bmatrix} 1 & 0 \\ 1 & \end{bmatrix}$$

14. $v1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$, $v2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$, $v3 = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$ **vektorlari chiziqli mustaqilmi? Agar bog'liq bo'lsa, birini boshqalari orqali ifodalang.**

[Manba: $ISALA$, hafta 1, mashq 14]

Yechim. $v_3 = \alpha v_1 + \beta v_2$ deb faraz qilamiz: komponentlar $\alpha = 1$(birinchi), $\beta = 1$(ikkinchi), $\alpha + \beta = 2$(uchinchi) — mos keladi. Demakv$_3 = v_1 + v_2$, no trivial bog'liqlik mavjud va to'plam bog'liq. Faqat ikkita mustaqil yo'nal mavjud; span — tekislik.

$$\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

15. $v_1 =, v_2 =$ **bo'lsin. (1) Ular** R^2 **ni hosil qiladimi?** $(2)b =$ **uchun**

$$21[]b_27$$

c_1, c_2 **ni toping shundayki** $c_1v_1 + c_2v_2 = b$**. (3) formuladan ifodalang. 4**

[Manba: $ISALA$, hafta 1, mashq 15]

Yechim. (1)–$(2)c_1 + 3c_2 = b_1, 2c_1 + c_2 = b_2$ tenglamalarini yechamiz. Chiqarib: $c_1 = b_1 - 3c_2, u$ holda $2b_1 - 6c_2 + c_2 = b_2$, demak $c_2 = (2b_1 - b_2)/5vac_1 = (-b_1 + 3b_2)/5$. Formulalar

har qanday (b_1, b_2) uchun aniqlangan, shuning uchun ha, ikkita vektor R^2 ni hosil qiladi. $(3) c_1 = (-7+12)/5 = 1, c_2 = (14-4)/5 = 2$: $b = v_1 + 2v_2$.

$$Q = uv = (2 : a = V) + 2Vo.$$

16. $u_1 =, u_2 =$ **bo'lsin. (1) Ular** R^2 **ni hosil qiladimi? (2) 2 3**

$$(1) u_2 = 6u_1, po$$

span**{**u_1, u_2 **} ni geometrik ravishda tasvirlang; (3) unga tegishlimi? −4 2**

[Manba: $ISALA$, hafta 1, mashq 16]

Yechim. Nisbatlarni solishtirish $-6/(-1) = 6va\frac{3}{2}$ **ko'rsatadiki, bosilgan juftlik proporsional emas, shuning uchun bosilgandek** uR^2 **ni qamrab oladi; mashqning standart versiyasida** $u2 = 3u1 = \begin{bmatrix} -3 \\ 6 \end{bmatrix}$ **— uni yechamiz.** $(1) u2 = 3u1$**, shuning uchun** span **— faqat** span**{**$u1$**}. (2) Geometrik jihatdan bu (−1, 2) yo'nalishidagi boshidan** o**'tgan yagona to'**g**'ri chiziq; ikki kollinear vektor hech qachon tekislikni qamrab olmaydi.** $(3) \begin{bmatrix} 8 \\ -4 \end{bmatrix}$ **uchun:** $-c = 8$ **va 2c** $= -4$**, ya'ni** $c = -8 vac = -2$ **— ziddiyat, demak yo'**q**;** $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ **uchun:** $-c = 1$ **va 2c** $= 2$**, ya'ni** $c = -1 vac = 1$ **— ziddiyat, demak yo'**q**.**

$$y = (\frac{1}{2}) \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} x + \begin{bmatrix} 1 \\ 0 \end{bmatrix} (16)$$

17. Faraz qilaylik $v1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$, $v2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$, $v3 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$**.** $(1) c1v1 + c2v2 + c3v3 = b$ **ni ixtiyoriy** $b = \begin{bmatrix} b1 \\ b2 \\ b3 \end{bmatrix}$ **uchun yeching; (2) nima uchun bu uch vektor** R^3 **ni qamrab olishini tushuntiring;** $(3) b = \begin{bmatrix} 4 \\ 2 \\ 6 \end{bmatrix}$ **ni ifodalang.**

$$c_1 = (b_1 + b_2 - b_3)/2$$

ixtiyoriyb $= b_2$ **uchun yeching; (2) uchta vektor** R^3 **ni egallaydigan sababini tushuntiring;**

$$c_1 = (b_1 + b_2 - b_3)/2$$

[Manba: $ISALA$, hafta 1, mashq 17]

Yechim. (1) Komponentlar: $c_1 + c_3 = b_1, c_2 + c_3 = b_2, c_1 + c_2 = b_3$. Uchtasini qo'shamiz: $2(c_1 + c_2 + c_3) = b_1 + b_2 + b_3$, demak $c_1 + c_2 + c_3 = {}^{b1+b2+b3}$. Keyin $c_1 = {}^{b1-b2+b3}, 22c_2 = {}^{-b1+b2+b3}$, $c_3 = {}^{b1+b2-b3}$. (2) Yechim istalgan o'ng tomon uchun mavjud va yagona, shuning uchun

span — butun R^3(vektorlar mustaqil). $(3)c_1 = ^{4-2+6} = 24, c_2 = ^{-4+2+6} = 2, c_3 = ^{4+2-6} = 0$:$b = 4v_1 + 2v_2$. 2 2

$$b = 4v_1 + 2v_2$$

18. Detalga uchta kuch ta'sir qiladi: $F_1 = N, F_2 = N, F_3 =$

$$\begin{bmatrix} 10 \\ 25 \\ -5 \end{bmatrix}$$

N. **(1) Rivojlantiruvchini toping; (2) detali muvozanatga keltiruvchiF_4 ni toping:**$F_1 + F_2 + F_3 + F_4 = 0$**.**

$$\begin{bmatrix} 40 \\ -15 \\ -20 \end{bmatrix}$$

[Manba: $ISALA$, hafta 1, mashq 18]

Yechim. (1) Komponent bo'yicha qo'shamiz: $F_1 + F_2 + F_3 =$

$$\begin{bmatrix} 10 \\ 25 \\ -5 \end{bmatrix}$$

$= 50N$. (2) Muvozanat uchun $F_4 = -(F_1 + F_2 + F_3) =$

$$\begin{bmatrix} -40 \\ 15 \\ 20 \end{bmatrix}$$

N kerak: bu rivojlantiruvchi vektorga aynan qarama-qarshi, shunda umumiy kuch nolga aylanadi.

2.3 Chiziqli algebra I, hafta 1: TA sessiya masalalari (alternativto'plam) — savollar va to'liq yechimlar

TA to'plami sakkizta qisqa masaladan iborat, har biri rasmiy yechimga ega; ikkalasini ham keltiramiz, taxminlarni batafsil ochib berib.

1. $a = (-1, 3, 2), b = (4, 0, -2)$**. Hisoblanga** $+b$ **va 3a** $-b$**;** a **vab** orasidagi** o**'rtacheni toping.**

[Manba: Zadachi TA, nedelya 1, zadacha 1]

Yechim. $a + b = (3, 3, 0)$; 3a $-b = (-3 - 4, 9 - 0, 6 + 2) = (-7, 9, 8)$. O'rta — o'rtacha $^{a+b} = (^3_2, 3_2, 0)$ — bu kesma o'rtasining skalyar formuladagi o'rtacha olish g'oyasi bilan bir xil.

2. Berilganp $+q = (5, 7)vap - q = (-1, 3)$**;**$pvaq$ **ni toping.**

[Manba: Zadachi TA, nedelya 1, zadacha 2]

Yechim. Ikkita tenglamani qo‘shamiz: 2p $= (4, 10)$, demakp $= (2, 5)$. Ayiramiz: 2q $= (6, 4)$, demakq $= (3, 2)$. Tekshirish: $(2, 5)+(3, 2) = (5, 7)va(2, 5)-(3, 2) = (-1, 3)$. «$Qo$‘shib ayirish» usuli — bu ikkita skalyar tenglamani bir vaqtda yechish vektor analogi.

3. au $= (1, 2, -1), v = (2, -1, 0)$**. Hisoblang** $-u+$ **3v; yozing cu** $+$ **dv komponent bo‘yicha; cu** $+dv(5, 0, -1)$ **ga teng bo‘lishi mumkinmi?**

[Manba: Zadachi TA, nedelya 1, zadacha 3]

Yechim. $-u+$ 3v $= (-1, -2, 1)+(6, -3, 0) = (5, -5, 1)$. Komponent bo‘yicha cu $+dv = (c+2d,$ 2c $-d, -c)$. (5, 0, −1) ga yetish uchun: uchinchi komponent $c = 1$ bo‘lishini talab qiladi; keyin $c+$ 2d $= 5d = 2$ beradi; va haqiqatan ham 2c $-d = 0$. Javob: ha,$u+$ 2v $= (5, 0, -1)$.

4. ar $= (2, -1), s = (-6, 3)$**. Barcha kombinatsiyalarni cr** $+ds$ **ni tasviring; vektorlar** R^2ni **to‘ldiradimi? Bog‘liqlik munosabatini toping.**

[Manba: TA masalalari, 1 hafta, 4-masala]

Yechim. $s = -$3r bo’lgani uchun, har qanday kombinatsiya $(c - 3d)r$ ko’rinishida bo’ladi — bu har doim r ning karrali. Span — bu orqali o’tgan to’g’ri chiziq: vektorlar tekislikni cho’zmaydi. 3r $+s = 0$ munosabati trivialemas, bu bog’liqlikni isbotlaydi (qarang (27)).

5. $v = (1, 0, 1), w = (1, 2, 0)$ **deb oling. cv** $+dw = (x, y, z)$ **yozing;** x, y, zni **bog’laydigan bitta tenglamani toping;** $b = (4, 6, 1)$ **ni tekshiring.**

[Manba: TA masalalari, 1 hafta, 5-masala]

Yechim. cv $+dw = (c + d,$ 2d, $c)$. O’qib: $z = cvay =$ 2d, demak $x = z+\frac{y}{2}$ — bu tekislikni tasvirlaydigan yagona chiziqli tenglama. uchun: o’ng tomon $= 1+\frac{6}{2} = 4 = x$, demak b tekislikda joylashgan; $z = 1, y = 6$ dan: $c = 1, d = 3$.

6. c, d **ni shunday topingki,** $c(1, 2) + d(3, -1) = (7, 3)$**; komponent bo’yicha tenglamalarni yozing va yeching.**

[Manba: TA masalalari, 1 hafta, 6-masala]

Yechim. Komponentlar: $c+$ 3d $= 7$ va 2c $-d = 3$. Ikkinchidan $d =$ 2c $- 3$; almashtiramiz: $c+$6c$-9 = 7$, demak 7c $= 16, c = \frac{16}{7}, d = 32_7 - 3 = 11_7$. Ikkala komponent ham aniq tekshiriladi.

7. $u = (1, 1, 0), v = (0, 1, 1), w = (1, 2, 1)$**.** w **yangi yo’nalish qo’shadimi?** span**{**u, v, w**}** **ni tasvirlang; nolga teng bo’lgan munosabatni keltiring.**

[Manba: TA masalalari, 1 hafta, 7-masala]

Yechim. $w = u+v$(komponent bo’yicha tekshirish: $1+0 = 1, 1+1 = 2, 0+1 = 1$), shuning uchun w ortiqcha. $u + v - w = 0$ munosabati trivialemas, demak uchta vektor bog’liq va span — bu faqat $uvav$(ular karrali emas) cho’zgan tekislik.

8. Ikkita vektor butun R^3ni **cho’zishi mumkinmi? Uchta vektor** R^3ni **cho’zishi mumkin va bog’liq bo’lishi mumkinmi? ”Yangi yo’nalishlar” orqali tushuntiring.**

[Manba: TA masalalari, 1 hafta, 8-masala]

Yechim. Ikkita vektor maksimal tekislikni cho’zadi — har bir vektor maksimal bitta yangi yo’nalish beradi, ikki yo’nalish bilan uch o’lchamli fazoni to’ldirib bo’lmaydi. Uchta bog’liq vektor ham maksimal tekislikni cho’zadi: bog’liqlik bittasining qolganlarining kombinatsiyasi ekanligini va uchinchi yo’nalish qo’shmasligini anglatadi. R^3 ga erishish uchun uchta mustaqil yo’nalish kerak; shuning uchun uchta bog’liq vektor hech qachon R^3ni cho’zmaydi.

3 2-Hafta — Skalyar va vektor ko'paytmalari,matritsalarga kirish (Lineal algebra I)

3.1 Lineal algebra I, 2-hafta: formulalar katalogi

2-hafta vektor geometriyasining ikkita ko'paytmasi va birinchi matritsa tushunchalarini qo'shadi. $u \cdot vu \cdot v = u_1v_1 + \cdots + u_nv_n = u^Tv$, $\cos\theta = \|u\|\|v\|$, $(45)proj_vu = u \cdot vv \cdot vv$, (46)

$$[u_2v_3 - u_3v_2]$$

$$u \times v = \begin{bmatrix} u2v3 & u3v2 \\ u3v1 & u1v3 \\ u1v2 & u2v1 \end{bmatrix}, \|u \times v\| = \|u\|\|v\|\|\sin\theta|(\textbf{yuza),}$$

$$u_1v_2 - u_2v_1$$

$(47)u \times v = -(v \times u), (u \times v) \perp u, v, (48)$

$$A = [a_{ij}] \in Rm \times n, (A + B)_{ij} = a_{ij} + b_{ij}, (cA)_{ij} = ca_{ij},$$

$(49)Ax = x_1a_1 + \cdots + x_na_n$(ustunlar), (50) bu yerda a_j — A matritsasining j-ustuni. O'lchamlar mos kelishi kerak: Ax faqat va faqat $Am \times no$'lchamda va $x \in R^n$ bo'lganda mavjud, natija R^m da joylashgan. Diagonal matritsa $D = diag(d_1, \ldots, d_n)har$ bir koordinatani masshtablaydi: $Dx = (d_1x_1, \ldots, d_nx_n)^T$.

3.2 Lineal algebra I, 2-hafta: ma'ruzadagi auditoriya masalalari — savollar va to'liq yechimlar

$$[3][]4$$

1. $u =, v =$**.** $u \cdot v$ **ni hisoblang; burchakni tasniflang.** -2 **5**

[Manba: LA ma'ruzasi, 2-hafta, 1-masala]

Yechim. $u \cdot v = 3 \cdot 4 + (-2) \cdot 5 = 12 - 10 = 2 > 0$**. Musbat skalyar ko'paytma** $\cos\theta > 0$ **degani, ya'ni burchak** o**'tkir.**

2. $u = \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix}, v = \begin{bmatrix} -1 \\ 3 \\ 0 \end{bmatrix}$ **bo'lsin:** $\|u\|vadist(u, v)$ **ni toping.**

[Manba: LA ma'ruzasi, 2-hafta, 2-masala]

Yechim. $\|u\| = \sqrt{(4 + 1 + 4)} = 3$. $u - v = \begin{bmatrix} 3 \\ -4 \\ 2 \end{bmatrix}$, demak $dist(u, v) = \sqrt{(9 + 16 + 4)} = \sqrt{(29)}$.

3. $u = \begin{bmatrix} 2 \\ k \\ -1 \end{bmatrix}$ **vektoriga perpendikular bo'ladigan** $v = \begin{bmatrix} 3 \\ 1 \\ 4 \end{bmatrix}$ **uchun** k **ni toping.**

[Manba: LA ma'ruzasi, 2-hafta, 3-masala]

Yechim. Perpendikulyarlik — bu $u \cdot v = 0$: $6 + k - 4 = 0$, demak $k = -2$.

$$Pemenne.OpToroHasIbHOCT'—3TO0U \cdot V = 0 : 6 + k - 4 = 0, 3HauUTk = -2.$$

4. $u = vav = o$**‘rtasidagi burchakni toping. 1 −1**

[Manba: Lektsiya LA, nedelya 2, zadacha 4]

Yechim. $u \cdot v = 1 - 1 = 0$: vektorlar perpendikulyar, $\theta = 90^\circ$ — formulalar kerak emas, skalyar ko‘paytma nolga teng bo‘lganda.

$$\cos\varphi = (u \cdot v)/(||u||||v||)$$

5. $u = \begin{bmatrix} 4 \\ 2 \end{bmatrix} niv = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ **ga proyeksiyaling.**

[Manba: Lektsiya LA, nedelya 2, zadacha 5]

Yechim. $u \cdot v = 6, v \cdot v = 2$**, shuning uchun** $proj_vu = (\frac{6}{2})v =$ **3v** $= \begin{bmatrix} 3 \\ 3 \end{bmatrix}$**. Qoldiq** $u-$ **3v** $= \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ **shartni qanoatlantiradi:** $(u - 3v) \cdot v = 0$ **— parallel va perpendikular qismlarga ajralish.**

6. $u = \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix}, v = \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix}$ **uchun** $u \times v$ **ni hisoblang va ortogonallikni tekshiring.**

$$2 \cdot 2 - 0 \cdot (-1)[4]$$

[Manba: Lektsiya LA, nedelya 2, zadacha 6]

Yechim. $(47)ga$ ko‘ra:$u \times v = 0 \cdot 3 - 1 \cdot 2 = -2$. Tekshirish: $1 \cdot 4 + 2 \cdot (-2) + 0 \cdot$

$$u \times v = \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix} \times \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix} = [0 \cdot 0 - 2 \cdot 2;$$

$$2 \cdot (-1) - 2 \cdot 0; 2 \cdot 2 - 0 \cdot (-1)] = \begin{bmatrix} -4 \\ -2 \\ 4 \end{bmatrix} (16)$$

$(-7) = 0va3 \cdot 4 + (-1)(-2) + 2 \cdot (-7) = 0$ — ikkala kirishga ham ortogonal. 2 1

7. Tomonlari $a = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}, b = \begin{bmatrix} 1 \\ 3 \\ -1 \end{bmatrix}$ **bo‘lgan parallelogramm yuzini toping.**

$$0 \cdot (-1) - 1 \cdot 3 - 3$$

[Manba: Lektsiya LA, nedelya 2, zadacha 7]

Yechim. $a \times b =$ **[**$0 \cdot (-1) - 1 \cdot 3$**;** $1 \cdot 1 - 2 \cdot (-1)$**;** $2 \cdot 3 - 0 \cdot 1$**]** $= \begin{bmatrix} -3 \\ 3 \\ 6 \end{bmatrix}$**, shuning uchun yuza**

$\|a \times b\| = \sqrt{9} + 9 + 36 = \sqrt{54} = 3\sqrt{6}$**.**

$$2 \cdot 3 - 0 \cdot 1 = 6$$

8. $A =$ **uchun** o**‘lchamni ko‘rsating,** a_{13}, a_{22} **ni toping va ustunlarni yozing** $035a_1, a_3$**.**

[Manba: Lektsiya LA, nedelya 2, zadacha 8]

Yechim. Ao‘lchami 2×3(avval qatorlar). $a_{13} = 4(1-$qator, 3-ustun$), a_{22} = 3.a_1 =, a_3 =$. 0 5

$$A = \begin{bmatrix} 2 & 4 & \\ 0 & 3 & 5 \end{bmatrix}$$

9. $A = \begin{bmatrix} 1 & -2 \\ 3 & 0 \end{bmatrix}, B = \begin{bmatrix} 4 & 1 \\ -1 & 5 \end{bmatrix}$ **bo‘lsin. 2A** $-B$ **ni toping.**

[Manba: Lektsiya LA, nedelya 2, zadacha 9]

Yechim. 2A $= \begin{bmatrix} 2 & -4 \\ 6 & 0 \end{bmatrix}$; element bo‘yicha ayiramiz: 2A $-B =$ [2 − 4, −4 − 1; 6 − (−1), 0 − 5] $= \begin{bmatrix} -2 & -5 \\ 7 & -5 \end{bmatrix}$.

10. Ao**‘lchami** $3 \times 2, B$ **—** $2 \times 3, x \in R^2, y \in R^3$**.** Ax, Ay, Bx, By **dan qaysilari aniqlangan?**

[Manba: Lektsiya LA, nedelya 2, zadacha 10]

Yechim. Ax: $(3 \times 2)(2 \times 1)OK$, natija R^3 da. By: $(2 \times 3)(3 \times 1)OK$, natija R^2 da. Ay mos kelmaydi $(2 = 3)vaBx$ mos kelmaydi $(3 = 2)$: ikkalasi ham aniqlanmagan. Faqat ichki o‘lchamlar mos kelishi kerak. 4

11. $D = diag(2, -1, 3), x = \begin{bmatrix} 4 \\ 5 \\ -2 \end{bmatrix}$ **bo’lsin.** $DxvaI$**3x ni toping.**

[Manba: Lektsiya LA, nedelya 2, zadacha 11]

Yechim. $Dx =$ **[**$2 \cdot 4$**;** $-1 \cdot 5$**;** $3 \cdot (-2)$**]** $= \begin{bmatrix} 8 \\ -5 \\ -6 \end{bmatrix}$**: har bir koordinata** $o’z$ **diagonal elementi bilan masshtablanadi.** I**3x** $= x = \begin{bmatrix} 4 \\ 5 \\ -2 \end{bmatrix}$**: birlik matritsa hech narsa qilmaydi.**

$$3 \cdot (-2) = -6$$

. $I_3 x = x = 5$: birlik matritsa hech narsa qilmaydi. -2

12. $A = \begin{bmatrix} 1 & 2 & 0 \\ -1 & 3 & 4 \\ 4 & 0 & 8 \end{bmatrix}, x = \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix}$ **bo’lsin.** Ax **ni ustunlar tasviri va qatorlar skalyar ko’paytmalari orqali hisoblang.**

$$\begin{bmatrix} 1 & 2 & 0 \\ -1 & 3 & 4 \\ 0 & 0 & 8 \end{bmatrix}$$

[Manba: Lektsiya LA, nedelya 2, zadacha 12]

Yechim. Ustunlar: $2 \cdot \begin{bmatrix} 1 \\ -1 \\ 4 \end{bmatrix} + (-1) \cdot \begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix} + 0 \cdot \begin{bmatrix} 0 \\ 4 \\ 8 \end{bmatrix} = \begin{bmatrix} 0 \\ -5 \\ 8 \end{bmatrix}$**. Qatorlar:** $\begin{bmatrix} 1 & 2 \end{bmatrix} \cdot x = 0$, $\begin{bmatrix} -1 & 3 \end{bmatrix} \cdot x = -5$, $\begin{bmatrix} 4 & 0 \end{bmatrix} \cdot x = 8$**; ustunma-ustun xuddi shu** $\begin{bmatrix} 0 \\ -5 \\ 8 \end{bmatrix}$**. Ikki tasvir har doim mos keladi.**

$$13. Chiziqli aylantirish L : R^3 \to R^3 L(x) Ax formulasi bilan aniqlanadi,$$
$$bunda A = [-130; 2 - 14; 0$$

12].

13. $A = va Ax =$ **berilgan;** x **ni toping. 3 −1 5**

[Manba: Lektsiya LA, nedelya 2, zadacha 13]

Yechim. Komponentlar: $x_1 + 2x_2 = 7, 3x_1 - x_2 = 5$. Ikkinchisidan $x_2 = 3x_1 - 5$;

$$x = [\frac{17}{7}]$$

almashtiramiz: $x_1 + 6x_1 - 10 = 7$, demak $7x_1 = 17, x_1 = \frac{17}{7}, x_2 = 16_7$. Shunday qilibx =. 16/7

14. Nol bo‘lmagan vektorlar $u + v va u - v$ **ortogonal.** $\|u\| va \|v\|$ **haqida nima deyish mumkin?**

[Manba: Lektsiya LA, nedelya 2, zadacha 14]

Yechim. Ortogonallik $0 = (u+v) \cdot (u-v) = \|u\|^2 - \|v\|^2$(kesishuvchi hadlar bekor qilinadi), bundan $\|u\| = \|v\|$: teng uzunliklar, vau vav — to‘g‘ri burchakli parallelogrammning yarim diagonallari.

3.3 Chiziqli algebra I, nedelya 2: mashqlar ISA — savollar va to‘liq yechimlar

1. $u = (-3, 2)^T, v = (4, 5)^T$ **bo‘lsin.** $u \cdot v$ **ni hisoblang; burchak** o**‘tkir, to‘**g**‘ri yoki** o**‘tkirmi?**

[Manba: $LAISA$, nedelya 2, mshr. 1]

Yechim. $u \cdot v = -12 + 10 = -2 < 0$, demak $\cos\theta < 0$: burchak o‘tkir.

2. aa $= (1, -2, 4)^T, b = (3, 0, -1)^T$**. Topinga** $\cdot b, \|a\|, \|b\|, \|a - b\|$**.**

[Manba: $ISALA$, hafta 2, mashq 2]

Yechim. $a \cdot b = 3 + 0 - 4 = -1$; $\|a\| = \sqrt{1} + 4 + 16 = \sqrt{21}$; $\|b\| = \sqrt{9} + 0 + 1 = \sqrt{10}$; $a - b = (-2, -2, 5)^T, \|a - b\| = \sqrt{4} + 4 + 25 = \sqrt{33}$.

3. $u = (2, k, 3)^T$ **vektori** $v = (-1, 4, 2)^T$ **vektoriga perpendikular bo'ladigan** k **ni toping.**

[Manba: $ISALA$, hafta 2, mashq 3]

Yechim. $u \cdot v = -2 + 4k + 6 = 0 \Rightarrow 4k = -4 \Rightarrow k = -1$.

4. $u = (3, 1)^T, v = (-2, 4)^T$ **uchun aniq** $\cos\theta ni$**, keyin** θ **ni gradus aniqligi bilan toping.**

[Manba: $ISALA$, hafta 2, mashq 4]

Yechim. $u \cdot v = 3 \cdot (-2) + 1 \cdot 4 = -6 + 4 = -2$**;** $\|u\| = \sqrt{10}, \|v\| = \sqrt{20} = 2\sqrt{5}$**;** $\cos\theta = -2/\sqrt{200} = -2/(10\sqrt{2}) = -\sqrt{2}/10 \approx -0.141$**, demak** $\theta \approx 98°$**.**

$$u \cdot v = 3 \cdot (-2) + 1 \cdot 4 = -6 + 4 = -2; ||u|| = \sqrt{1}0,$$
$$||v|| = \sqrt{2}0 = 2\sqrt{5}; \cos\theta = (u \cdot v)/(||u|| \cdot ||v||) =$$

$$(-2)/(\sqrt{1}0 \cdot 2\sqrt{5}) = (-1)/(\sqrt{5}0) = (-1)/(5\sqrt{2}) = -\sqrt{\frac{2}{10}}$$

$-\frac{1}{2} \approx -0.1414$, shuning uchun $\theta \approx 98^\circ$. $\sqrt{5}$

5. $u = (6, 1)^T$ **vektorining** $v = (2, -1)^T$ **vektoriga proyeksiyasini toping;** $u - proj$ **ning** v **ga perpendikularligini tekshiring.**

[Manba: $ISALA$, hafta 2, mashq 5]

Yechim. $u \cdot v = 12 - 1 = 11, v \cdot v = 5$: $proj_v u = \frac{11}{5}(2, -1)^T = (22_5, -\frac{11}{5})^T$. Qoldiq: $u - proj = (\frac{8}{5}, 16_5)^T$; v bilan skalyar ko'paytma: $16_5 - \frac{16}{5} = 0$. Tekshirildi.

6. Nolga teng bo'lmagan p, q **vektorlari** $p \cdot q = \|p\|\|q\|$ **shartni qanoatlantiradi. Ular yo'nalishi haqida qanday xulosa chiqarish mumkin?**

[Manba: $ISALA$, hafta 2, mashq 6]

Yechim. $(45)dan, \cos\theta = 1$, demak $\theta = 0$: vektorlar bir xil yo'nalishda (biri ikkinchisining musbat karrali). Burchak formulasidagi tenglik Koshi-Bunykovskiy-Shvarts g'oyasining chegaraviy holati.

7. $a = (2, 1, -1)^T, b = (0, 3, 2)^T$ **uchun** $a \times bvab \times a$ **ni hisoblang; belgilar munosabatini tekshiring.**

$$[2 \cdot 3 - (-1) \cdot 0; 5 - 5]$$

[Manba: $ISALA$, hafta 2, mashq 7]

Yechim. $a \times b =$ **[**$1 \cdot 2 - (-1) \cdot 3$**;** $(-1) \cdot 0 - 2 \cdot 2$**;** $2 \cdot 3 - 1 \cdot 0$**]** $= \begin{bmatrix} 5 \\ -4 \\ 6 \end{bmatrix}$**, va** $b \times a = \begin{bmatrix} -5 \\ 4 \\ -6 \end{bmatrix} = -(a \times b)$

— ishora munosabatini tasdiqlaydi.

$$[2 \cdot 3 - 1 \cdot 0; 1 \cdot 0 - (-1) \cdot 3; (-1) \cdot 3 - 2 \cdot 2] = \begin{bmatrix} 6 \\ 3 \\ -7 \end{bmatrix}$$

$(48)ga$ mos keladi.

8. Mashq 7 dan olingan $a \times b$ **vektorining** $avab$ **ikkalasiga ham perpendikular ekanligini tekshiring.**

[Manba: $ISALA$, hafta 2, mashq 8]

Yechim. $a\cdot(a\times b)=2\cdot 5+1\cdot(-4)+(-1)\cdot 6=10-4-6=0$;$b\cdot(a\times b)=0\cdot 5+3\cdot(-4)+2\cdot 6=-12+12=0$. Ikkalasi ham nolga teng, (47) dan keyin kafolatlanganidek.

9. $u=(1,2,2)^T, v=(3,-1,0)^T$ **tomonli parallelogrammning yuzini toping.**

$$[2\cdot 0-2\cdot(-1);2\cdot 3-1\cdot 0;1\cdot(-1)-2\cdot 3]$$

; yuz $=\sqrt{4}+36+49=\surd$

[Manba: $ISALA$, hafta 2, mashq 9]

Yechim. $u\times v=$ **[**$2\cdot0-2\cdot(-1)$**;** $2\cdot3-1\cdot0$**;** $1\cdot(-1)-2\cdot3$**]** $=\begin{bmatrix}2\\6\\-7\end{bmatrix}$**, norma** $=\sqrt{4}+36+49=\sqrt{89}$**.**

$$||u\times v||=\sqrt{((2\cdot 0-2\cdot(-1))^2+(2\cdot 3-1\cdot 0)^2+(1\cdot(-1)-2\cdot 3)^2)}=\sqrt{(4+36+49)}=\sqrt{8}9$$

10. $P=(1,0,1), Q=(3,1,0), R=(0,2,4)$**:** PQR **uchburchakning yuzini toping.**

[Manba: $ISALA$, hafta 2, mashq 10]

Yechim. P dan chiqadigan tomonlar: $PQ=(2,1,-1)^T, PR=(-1,2,3)^T$. Vektor ko'paytmasi: $PQ\times PR=(1\cdot 3-(-1)\cdot 2;(-1)\cdot(-1)-2\cdot 3;2\cdot 2-1\cdot(-1))=(5,-5,5)$, norma

$$\sqrt{(25+25+25)}=\sqrt{(75)}=5\sqrt{(3)}$$

; uchburchak yuzi $=(\frac{1}{2})\cdot 5\sqrt{3}=5\sqrt{3}/2$(parallelogramm yuzining yarmi). $uvav$ ni almashtirish vektor ko'paytmasining ishorasini teskarisiga o'zgartiradi, yuz o'zgarmaydi.

11. $a=(1,t,2)^T vab=(3,6,6)^T$ **vektorlari** $a\times b=0$ **shartni qanoatlantiradigan barcha haqiqiy** t **larni toping; agar bunday** $tlar$ **bo'lmasa, tushuntiring.**

[Manba: $ISALA$, hafta 2, mashq 11]

Yechim. Parallel vektorlar nol vektor ko'paytmasini beradi, demak $a=cb$ bo'lishi kerak: $1=3c\Rightarrow c=\frac{1}{3}$; unda $t=6c=2$; uchinchisini tekshiramiz: $2=6c=2$ — mos keladi. Shunday qilib $t=2$ yagona qiymat (unda $a=\frac{1}{3}b$).

12. $a=(1,0,2)^T, b=(2,1,-1)^T, c=(0,3,1)^T$**.** $a\cdot(b\times c)$ **ni hisoblang; nol qiymatini talqin qiling.**

$$1\cdot 1-(-1)\cdot 34$$

[Manba: $ISALA$, hafta 2, mashq 12]

Yechim. $b\times c=$ **[**$1\cdot1-(-1)\cdot3$**;** $(-1)\cdot0-2\cdot1$**;** $2\cdot3-1\cdot0$**]** $=\begin{bmatrix}4\\-2\\6\end{bmatrix}$**;** $a\cdot(b\times c)=1\cdot4+0\cdot(-2)+2\cdot6=16$**.**

$$\begin{bmatrix}1&0&2\\2&1&-1\\0&3&1\end{bmatrix}$$

Bu uch vektorga tortilgan parallelepipedning belgilangan hajmi; nol qiymati komplanarlikni ($3D$ hajmi yo'qligini) anglatardi. Bu yerda $|16|=16$ kub birlik. 2 −1 0

13. $A = \begin{bmatrix} 2 & -1 & 0 \\ 4 & 3 & 5 \end{bmatrix}$ **matritsasi berilgan.** O**'lchamini ko'rsating;** a_{12}, a_{23} **elementlarini toping; har bir qator va ustunni vektor sifatida yozing.**

[Manba: $ISALA$, hafta 2, mashq 13]

Yechim. O'lchami 2×3(avval qatorlar). $a_{12} = -1(1-$qator, 2-ustun$), a_{23} = 5$. Qatorlar: $\begin{bmatrix} 2 & -1 & 0 \end{bmatrix}, \begin{bmatrix} 4 & 3 & 5 \end{bmatrix}$; ustunlar: $\begin{bmatrix} 2 \\ 4 \end{bmatrix}, \begin{bmatrix} -1 \\ 3 \end{bmatrix}, \begin{bmatrix} 0 \\ 5 \end{bmatrix}$.

$$[1][]35 - 1$$

14. $A = \begin{bmatrix} 1 & 3 \\ -2 & 4 \end{bmatrix}, B = \begin{bmatrix} 5 & -1 \\ 0 & 2 \end{bmatrix}$ **matritsalar berilgan.** $A + B, A - B, -$**2A, 3A − 2B ni hisoblang.**

[Manba: $ISALA$, hafta 2, mashq 14]

Yechim. $A + B = \begin{bmatrix} 6 & 2 \\ -2 & 6 \end{bmatrix}$; $A - B = \begin{bmatrix} -4 & 4 \\ -2 & 2 \end{bmatrix}$; $-$2A $= \begin{bmatrix} -2 & -6 \\ 4 & -8 \end{bmatrix}$; 3A $-$ 2B $=$ [$3 - 10, 9 + 2$; $-6 - 0$, $12 - 4$] $= \begin{bmatrix} -7 & 11 \\ -6 & 8 \end{bmatrix}$.

15. A **matritsasi** $4 \times 3$$o$**'lchamda,** B **—** $3 \times 2, x \in R^2, y \in R^3, z \in R^4$**.** Ax, Ay, Az, Bx, By, Bz **dan qaysilari aniqlangan va ularning chiqish** o**'lchamlari qanday?**

[Manba: $ISALA$, hafta 2, mashq 15]

Yechim. Ay: $(4 \times 3)(3 \times 1)$ — aniqlangan, natija R^4. Bx: $(3 \times 2)(2 \times 1)$ — aniqlangan, natija R^3. Qolganlari (Ax, Az, By, Bz) ichki o'lchamlari mos kelmaydi va aniqlanmagan.

16. $D = diag(-2, 3, 1), x = (4, -1, 5)^T$ **berilgan.** Dx **ni hisoblang;** $Dhar$ **bir koordinataga nima qilishini tushuntiring.**

$$Dx = \begin{bmatrix} -2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix}, x = (2, -1, 3)^T.$$

[Manba: $ISALA$, hafta 2, mashq 16]

Yechim. Dx**: birinchi koordinata −2 ga masshtablanadi** $(aks$ **etish bilan cho'zish), ikkinchi —** $3ga$**, uchinchi — 1/2 ga.**

17. $A =$ **matritsasi berilgan.** $Axni(a)$ **ustunlarning 4 0 −1 kombinatsiyasi sifatida,** (b) **qatorlarning skalyar ko'paytmalari orqali hisoblang.**

$$1 - 22 + 2 + 9 = 13$$

[Manba: $ISALA$, hafta 2, mashq 17]

Yechim. $(a)2 - 1 + 3 =$

$$2 \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix} - \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} + \begin{bmatrix} 3 \\ 0 \\ 3 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \\ 9 \end{bmatrix}$$

. (b)1-qator: $1 \cdot 2 + (-2)(-1) + 3 \cdot 3 = 13$; 2-qator: $4 \cdot 2 + 0 \cdot (-1) + (-1) \cdot 3 = 5$; 3-qator: $2 \cdot 2 + 2 \cdot (-1) + 2 \cdot 3 = 6 - 2 + 6 = 10$.

$$\begin{bmatrix} 1 & 4 & 2 \\ -2 & 0 & 2 \\ -1 & -1 & 2 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} = \begin{bmatrix} 13 \\ 5 \\ 10 \end{bmatrix}$$

Xuddi shu natija, bo'lishi kerakgidek.

18. $a_1 = (1,2)^T, a_2 = (3,-1)^T$ **vektorlar berilgan.** $x_1a_1 + x_2a_2 = (11,1)^T$ **tenglamani qanoatlantiruvchi** $x = (x_1, x_2)^T$ **ni toping.**

[Manba: $ISALA$, hafta 2, mashq 18]

Yechim. Komponentlar: $x_1 + 3x_2 = 11, 2x_1 - x_2 = 1$. Ikkinchisidan $x_2 = 2x_1 - 1$; keyin $x_1 + 6x_1 - 3 = 11 \Rightarrow 7x_1 = 14 \Rightarrow x_1 = 2, x_2 = 3$. 2 4

19. $A = \begin{bmatrix} 2 & 4 \\ -1 & -2 \end{bmatrix}$ **matritsasi berilgan.** $Ax = 0$ **tenglamani qanoatlantiruvchi barcha** x **ni tavsiflang; nima uchun ustunlar chiziqli bogliq?**

[Manba: $ISALA$, hafta 2, mashq 19]

Yechim. $2x_1 + 4x_2 = 0 va - x_1 - 2x_2 = 0$ — bir xil tenglama: $2x_1 = -4x_2$. Yechimlar: $x = t \cdot \begin{bmatrix} -2 \\ 1 \end{bmatrix}, t \in R$ — butun yechimlar to'g'ri chiziq. Trivial bo'lmagan kombinatsiya $-2 \cdot a_1 + 1 \cdot a_2 = 0$ mavjud, bu ustunlarning chiziqli bog'liqligidir (2-ustun 1-ustunning -2 barobariga teng).

$$[][]1 = (2,-1,4)_T va A0 = (3,2,0)_T . Aniyozing$$

20. A **matritsasi** $3 \times 2 o$**'lchamda,** $A \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix} va A \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ 0 \end{bmatrix}$**.** A **ni yozing va** $A \begin{bmatrix} -2 \\ 5 \end{bmatrix}$ **ni qator skalyar ko'paytmalarisiz hisoblang.**

$$A = \begin{bmatrix} 2 & 3 \\ -1 & 2 \\ 5 & 4 \end{bmatrix}$$

[Manba: $ISALA$, hafta 2, mashq 20]

Yechim. Ikkita tasvir — bu A ning ustunlari: $A = -12$. Chiziqlilik: $A = 540$

$$\begin{bmatrix} 2 & 3 \\ 5 & 11 \end{bmatrix}$$

$-2Ae_1 + 5Ae_2 = -2 - 1 + 52 = 12$. 4 0 −8

21. Bir o**'lchamli vektorlar uchun** $(u+v) \cdot (u-v)$ **ni oching;** $u + v \perp u - v$ **bo'lishi uchun zarur va yetarli shart** $\|u\| = \|v\|$ **ekanligini isbotlang.**

[Manba: $ISALA$, hafta 2, mashq 21]

Yechim. Taqsimlab: $(u+v) \cdot (u-v) = u \cdot u - u \cdot v + v \cdot u - v \cdot v = \|u\|^2 - \|v\|^2$ (o'rtadagi hadlar bekor qilinadi). Perpendikulyarlik bu nolga teng bo'lishini anglatadi, ya'ni $\|u\|^2 = \|v\|^2$, va normalar manfiy bo'lmagani uchun $\|u\| = \|v\|$.

22. $O'z$**-**o**'zini tekshirish: (1) skalyar ko'paytma va vektor ko'paytma qaysi ob'ektni beradi?** $(2) u \cdot v$ **belgisi burchakni qanday tasniflaydi? (3) Nima uchun**

2×3 **matritsasi** R^3 **dan vektorni ko'paytiradi, lekin** R^2 **dan emas?** $(4)Axdax$ **komponentlari qanday ishlaydi? (5) 2-haftadagi qaysi amallar teng** o**'lchamli vektlarni talab qiladi, qaysilari aynan** R^3 **dan vektlarni?**

[Manba: $LAISA$, 2-hafta, $o'z$-o'zini tekshirish]

Yechim. (1) Skalyar; vektor (R^3da). (2) Musbat $\Rightarrow$ o'tkir, nol $\Rightarrow$ to'g'ri, manfiy $\rightarrow$ o'tmas. $(3)Ax$ ko'paytmasi ichki o'lchamlarning mosligini talab qiladi: $2 \times 3va3 \times 1$ ishlaydi; $2 \times 1esa$ yo'q. (4) Chiziqiy kombinatsiyadagi og'irliklar kabi $x_1a_1 + \cdots + x_na_n$. (5) Skalyar ko'paytma, qo'shish va ayirish teng o'lchamli vektorlarni talab qiladi; vektor ko'paytmasi va uning geometriyasi (yuzalar, normalar) faqat R^3 da mavjud.

3.4 Chiziqli algebra I, 2-hafta: TA sessiyasi (15 masala) — savollarva to'liq yechimlar

$$[][] - 25$$

1.$a =, b =$**: hisoblanga** $\cdot b$**; burchakni tasniflang. 4 1**

[Manba: $LATA$, 2-hafta, masala 1]

Yechim. $a \cdot b = (-2) \cdot 5 + 4 \cdot 1 = -10 + 4 = -6 < 0$**:** o**'tmas (botiq).**

2. $v = \begin{bmatrix} -2 \\ 1 \\ 2 \end{bmatrix}$ **vektorining normasini va birlik vektorni toping.**

[Manba: $LATA$, 2-hafta, masala 2]

Yechim. $\|v\| = \sqrt{4} + 1 + 4 = 3$**; birlik vektor** $\hat{v} =$ **[−2/3; 1/3; 2/3].**

3. $p = \begin{bmatrix} -2 \\ 4 \\ 3 \end{bmatrix} vaq = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$ **orasidagi masofani toping.**

[Manba: $LATA$, 2-hafta, masala 3]

Yechim. $p - q = \begin{bmatrix} -2 & 2 \\ 4 & 1 \\ 3 & 0 \end{bmatrix} = \begin{bmatrix} -4 \\ 3 \\ 3 \end{bmatrix}$**, masofa** $= \sqrt{16} + 9 + 9 = \sqrt{34}$**.**

4. Agar $\begin{bmatrix} 2 \\ t \\ 4 \end{bmatrix}$ **vektori** $\begin{bmatrix} 1 \\ -3 \\ 2 \end{bmatrix}$ **ga perpendikulyar bo'lsa,** t **ni toping.**

[Manba: $LATA$, 2-hafta, masala 4]

Yechim. $2 - 3t + 8 = 0 \Rightarrow t = \frac{10}{3}$.

$$\begin{bmatrix} 2 \\ -1 \end{bmatrix} \cdot \begin{bmatrix} t \\ -3 \end{bmatrix} = 0 \Rightarrow 2t + 3 = 0 \Rightarrow t = -\frac{3}{2}$$

5. $u =, v =$ **uchun** $\cos\theta$ **ni toping. 1 3**

$$\cos\theta = 1/\sqrt{50} = \sqrt{\frac{2}{10}}$$

[Manba: $LATA$, 2-hafta, masala 5]

Yechim. $u \cdot v = 2 \cdot (-1) + 1 \cdot 3 = 1$**;** $\|u\| = \sqrt{5}, \|v\| = \sqrt{10}$**;** $\cos\theta = 1/\sqrt{50} = \sqrt{2}/10 \approx 0.141, \theta \approx 82$**°.**

6. $u = (5,-1)^T niv = (2,1)^T$ **ga proyeksiya qiling;** u **ni parallel +** **perpendikulyar qismlarga yozing.**

$$[\frac{7}{5}; \frac{18}{5}]$$

[Manba: $LATA, 2$-hafta, masala 7]

Yechim. $u \cdot v = 9, v \cdot v = 5$: $proj =$

$$(\frac{9}{5}) \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

. Perpendikulyar qism:

$$\begin{bmatrix} 5 \\ -1 \end{bmatrix} - [\frac{14}{5}; \frac{9}{5}]$$

$u - proj =$

$$[\frac{11}{5}; -\frac{18}{5}]$$

. Yoyilma:

$$u = [\frac{14}{5}; \frac{9}{5}] + [\frac{11}{5}; -\frac{18}{5}]$$

.

7. $a = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}, b = \begin{bmatrix} 0 \\ 4 \\ 1 \end{bmatrix}$ **uchun** $a \times b$ **ni hisoblang.**

$$(-1) \cdot 1 - 3 \cdot 4 - 13$$

[Manba: $LATA, 2$-hafta, masala 8]

Yechim. $a \times b =$ **[**$(-1) \cdot 1 - 3 \cdot 4$**;** $3 \cdot 0 - 2 \cdot 1$**;** $2 \cdot 4 - (-1) \cdot 0$**]** $= \begin{bmatrix} -13 \\ -2 \\ 8 \end{bmatrix}$**.**

$$|-48; 0-1|$$

8. $A = (0,0,0), B = (2,1,0), C = (1,0,3)$**: uchburchak yuzasini toping.**

$$\begin{bmatrix} 2 & 1 & 0 \\ 1 & 0 & 3 \end{bmatrix}$$

[Manba: $LATA, 2$-hafta, masala 9]

Yechim. $AB = AC =$ vektor ko’paytmasi

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}^{1/2}$$

$= 0 \cdot 1 - 2 \cdot 3 = -6$, norma $= 46$; yuzasi $= \frac{46}{2}$.

$$S = (\frac{1}{2}) \cdot ||(2;1;0) \times (1;0;3)|| = (\frac{1}{2}) \cdot ||(3;-6;$$
$$-1)|| = (\frac{1}{2}) \cdot \sqrt{(}3^2 + (-6)^2 + (-1)^2) = (\frac{1}{2}) \cdot \sqrt{46}$$

9. $M =$ **uchun** o**'lcham,** m_{21}**, barcha ustunlar. 4 3 5**

$$\begin{bmatrix} 1 & 0 & -2 \\ 4 & 3 & 5 \end{bmatrix}$$

[Manba: $LATA, 2$-hafta, masala 10]

Yechim. 2×3; $m_{21} = 4$; ustunlar„. 4 3 5

$$\begin{bmatrix} 2 & 1 & 0 \\ 0 & 4 & 3 \\ 5 & 0 & 0 \end{bmatrix}$$

10. $A = \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 4 \\ 2 & -2 \end{bmatrix}$**. 3A + 2B ni toping.**

[Manba: $LATA, 2$-hafta, masala 11]

Yechim. 3A $= \begin{bmatrix} 6 & 3 \\ -3 & 9 \end{bmatrix}$, 2B $= \begin{bmatrix} 0 & 8 \\ 4 & -4 \end{bmatrix}$; yig'indi $= \begin{bmatrix} 6 & 11 \\ 1 & 5 \end{bmatrix}$.

11. $A = \begin{bmatrix} 2 & -1 & 3 \\ 0 & 4 & 1 \end{bmatrix}$, $x = \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}$**.** Ax **ni ustunlar kombinatsiyasi sifatida hisoblang.**

[Manba: $LATA, 2$-hafta, masala 12]

Yechim. $1 - 2 + 3 ==$.

$$\begin{bmatrix} 0 & 4 & 1 \\ -2 & 0 & 4 \\ 1 & 3 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = x \begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix} + y \begin{bmatrix} 4 \\ 0 \\ 3 \end{bmatrix} + z \begin{bmatrix} 1 \\ 4 \\ 2 \end{bmatrix} = [0x + 4y + 1z;$$
$$-2x + 0y + 4z;$$

$$1x + 3y + 2z] = \begin{bmatrix} 4y & + & z & \\ -2x & + & 4z & \\ x & + & 3y & + & 2z \end{bmatrix} = \begin{bmatrix} 9 \\ 5 \\ 13 \end{bmatrix}$$

$$\begin{bmatrix} 2 & -1 \\ 3 & 2 \end{bmatrix} + \begin{bmatrix} 2 \\ 2 \end{bmatrix} + \begin{bmatrix} 9 \\ 13 \end{bmatrix}$$

12. $a_1 =, a_2 =, b =$**. [**$a_1 a_2$**]**$x = b$ **bo'ladigan barcha** x **larni toping. 2 4 6**

[Manba: $LATA, 2$-hafta, masala 13]

Yechim. Ikkala tenglama ham $x_1 + 2x_2 = 3$ ga kelib chiqadi (tizma mos, lekin

$$\begin{bmatrix} 3 & & 2t \\ -7 & 2 & \\ -9 & 4 & \end{bmatrix}$$

ortiqcha, chunki $a_2 = 2a_1 vab = 3a_1)$. Umumiy yechim:$x =, t \in R$ — $tbir$ xil kombinatsiya uchun cheksiz ko'p og'irliklar.

13. Koordinatalarni ochmasdan, nima uchun $(u \times v) \cdot (u + v) = 0$ **ekanini tushuntiring.**

[Manba: $LATA, 2$-hafta, masala 14]

Yechim. Vektor ko'paytmasi ikkala kirish vektoriga ham perpendikulyar, shuning uchun ularning har qanday kombinatsiyasiga perpendikulyar: $(u \times v) \cdot u = 0 va (u \times v) \cdot v = 0$; qo'shib, $u + v$ bilan skalyar ko'paytma $0 + 0 = 0$ ga teng.

$$[u_1 u_2 u_3] \times [v_1 v_2 v_3] \cdot ([u_1 u_2 u_3] + [v_1 v_2 v_3]) = 0$$

14. Ao**'lchami** 2×2**, bunda** $A \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \end{bmatrix} vaA \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$**.** $AvaA \cdot \begin{bmatrix} 5 \\ -2 \end{bmatrix}$ **ni toping.**

[Manba: $TALA, 2$-hafta, masala 15]

Yechim. Standart bazis vektorlarining tasvirlari — ustunlar:

$$A = \begin{bmatrix} 2 & 3 \\ -1 & 4 \end{bmatrix}$$

. U holda $A \cdot \begin{bmatrix} 5 \\ -2 \end{bmatrix} = 5 \cdot \begin{bmatrix} 2 \\ -1 \end{bmatrix} - 2 \cdot \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 10 & 6 \\ -5 & 8 \end{bmatrix} = \begin{bmatrix} 4 \\ -13 \end{bmatrix}$; chiziqlilik hisobni skalyar ko'paytmasiz ikki masshtablashga aylantiradi.

4 3-hafta — Matritsa algebrasi, determinantlar, aralashko'paytma (Chiziqli algebra I)

4.1 Chiziqli algebra I: 3-hafta formulalari katalogi

$$(A + B)_{ij} = a_{ij} + b_{ij}, (cA)_{ij} = ca_{ij}, (A^T)_{ij} = a_{ji},$$
$$(A^T)^T = A, (65)(A + B)^T = A^T + B^T, (cA)^T = cA^T, A^T$$

$$= A(simmetrik),$$

(66)

$$A \in R^{m\times n}, B \in R^{n\times p} \Rightarrow AB \in R^{m\times p}, (67)$$

(67)

$$(AB)_{ij} = \Sigma_{k=1}^{n} a_{ik} b_{kj}, (68)$$

$AB = [Ab_1 \cdots Ab_p]$ (ustunlar qoidasi), $(71) u^T v$ = skalyar, uv^T = matritsa (tashqi ko'paytma). $(72) ab \det = ad - bc$(belgili maydon), $(73) cd C_{ij} = (-1)^{i+j} \det M_{ij}$(kofaktor),

$$C_{ij} = (-1)^{i+j} \det M_{ij}, \det A = a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13}, (74)$$

bu yerda M_{ij} — minor ($i-$qator va j-ustun chiqarib tashlangan). Ixtiyoriy qator yoki ustun bo'yicha ajratish ruxsat berilgan; nollardan iborat qatorlar va ustunlar arifmetikani soddalashtiradi. Uchburchak matritsalar uchun determinant diagonall elementlari ko'paytmasiga teng. Elementar qatorlarni o'zgartirish qoidalari jadval 1-da keltirilgan. Jadval 1: Kvadrat matritsalar determinantlari qoidalari (qatorlar uchun versiyalar; ustunlar uchun versiyalar ham to'g'ri).

A ustidagi operatsiya	$\det Aga$ ta'siri
Ikkita qatorni almashtirish Bitta qatorni cga ko'paytirish	belgini o'zgartirir determinantni cga ko'paytirir o'zgarmaydi determinant $0 \det(A^T) = \det A$
Boshqa qatordan ko'paytmani qo'shish Nol qator yoki ikki	
proporsional qatorlar Transponirlash Bir xil o'lchamdagi	
kvadrat matritsalar ko'paytmasi $\det(AB) = \det A \det B$	

Butun $n \times n$ matritsasini masshtablashtirish barcha n qatorlarni masshtablaydi: $\det(cA) = c^n \det A$. Determinantlar qo'shishga nisbatan taqsimlanmaydi: $A = B = I_2$ bo'lganda, $\det(A+B) = 4, \det A + \det B = 2$ esa. $(u\times v)\cdot w = u\cdot(v\times w) = \det \begin{bmatrix} u & v & w \end{bmatrix} (S.K)$, $(75) h = |(u \times v) \cdot w| V_{parallelepiped} = |(u \times v) \cdot w|, , (76) \|u \times v\| V_{tetrahedron} = \frac{1}{6} |\det \begin{bmatrix} u & v & w \end{bmatrix}|, T = 0 \Leftarrow\Rightarrow u, v, w$ komplanar. (77)

4.2 Chiziqli algebra I: 3-hafta ma'ruzasi masalalari (15) — savollarva to'liq yechimlar

$$[][2]1 - 21$$

1. $A = \begin{bmatrix} 1 & -2 \\ 3 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 1 \\ -1 & 4 \end{bmatrix}$ **bo'lsin. 2A** $-B$ **ni hisoblang va (2, 1) elementini ko'rsating;** $Ani 2 \times 3$ **matritsasiga qo'shish mumkinmi?**

[Manba: LA ma'ruzasi, 3-hafta, masala 1]

Yechim. $\text{2A} = \begin{bmatrix} 2 & -4 \\ 6 & 0 \end{bmatrix}$; $\text{2A} - B = [2 - 2, -4 - 1; 6 - (-1), 0 - 4] = \begin{bmatrix} 0 & -5 \\ 7 & -4 \end{bmatrix}$; (2, 1) elementi $2(3) - (-1) = 7$. 2×3 matritsasi bilan qo'shish aniqlanmagan — ikkala o'lcham ham mos kelishi kerak, umumiy elementlar soni emas.

2. $C = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 3 & 2 \end{bmatrix}$ **bo'lsin.** C^T **ni uning** o**'lchami bilan yozing;** CC^T **ni hisoblang;** u **simmetrikmi?**

[Manba: LA ma'ruzasi, 3-hafta, masala 2]

Yechim. $C^T = \begin{bmatrix} 1 & 0 \\ 2 & 3 \\ 1 & 2 \end{bmatrix}$ (3×2). $CC^T = [1+4+1, 0+6-2;\ 0+6-2, 0+9+4] = \begin{bmatrix} 6 & 4 \\ 4 & 13 \end{bmatrix}$. Diagonall elementlar — o'zlari bilan skalyar ko'paytirilgan qatorlar; tashqi diagonallari teng — matritsa simmetrik.

3. Yuqoridagi $Cvax = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$ **uchun** Cx **ni ikki usul bilan hisoblang.**

[Manba: LA ma'ruzasi, 3-hafta, masala 3]

Yechim. Qatorlar: $1\cdot2+2\cdot(-1)+(-1)\cdot1 = -1$**;** $0+3\cdot(-1)+2\cdot1 = -1$**; demak** $Cx = \begin{bmatrix} -1 \\ -1 \end{bmatrix}$**.**

Ustunlar: $2\cdot\begin{bmatrix}1\\0\end{bmatrix} + (-1)\cdot\begin{bmatrix}2\\3\end{bmatrix} + 1\cdot\begin{bmatrix}-1\\2\end{bmatrix} = \begin{bmatrix}-1\\-1\end{bmatrix}$**.** O**'lchamlar** $(2\times3)(3\times1)2\times1$ **chiqish beradi.**

$$remenne.CTpoKH : 1-2+2-(-1)+(-1)-1 = -1; 0+0-(-1)+2 = -1;$$

$$3HAUYUTCOX = [1] : \begin{bmatrix} 17 & 9 & \\ a & | & P14 \end{bmatrix}$$

Ustunlar: $2 - + =$. O'lchamlar $(2\times3)(3\times1)$ chiqish 2×1 beradi.

$$C = \begin{bmatrix} 0 & 3 & 2 \\ -1 & 0 & 0 \\ 0 & -1 & 0 \end{bmatrix}$$

4. 1-masaladagi A, B **uchun** $ABvaBA$ **ni hisoblang. Tenglar?**

$$\begin{bmatrix} 5 & 1 & 2 \\ -2 & -1 & 1 \\ 1 & -2 & 4 \end{bmatrix}$$

[Manba: LA ma'ruzasi, 3-hafta, masala 4]

Yechim. $AB =$

$$\begin{bmatrix} 6 & 3 \\ 8 & 7 \end{bmatrix}$$

, $BA =$

$$\begin{bmatrix} 8 & 10 \\ 0 & 0 \end{bmatrix}$$

. Turlicha; hatto ikkala tartibda ham mavjud bo’lganda, tenglik kafolatlanmaydi.

$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 1 & 1 \end{bmatrix},$$

$$D = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

5. $C = \begin{bmatrix} 1 & 2 & -1 \\ 0 & 3 & 2 \end{bmatrix}$, $D = \begin{bmatrix} 2 & 0 \\ 1 & -1 \\ 3 & 2 \end{bmatrix}$ **bo’lsin.** $CDvaDCo$**’lchamlarini oldindan aytib, ikkalasini hisoblang.**

[Manba: LA ma’ruzasi, 3-hafta, masala 5]

Yechim. CD: $(2 \times 3)(3 \times 2) = 2 \times 2$; DC: $(3 \times 2)(2 \times 3) = 3 \times 3$. $CD = [2 + 2 - 3, 0 - 2 - 2;$ $0 + 3 + 6, 0 - 3 + 4] = \begin{bmatrix} 1 & -4 \\ 9 & 1 \end{bmatrix}$; $DC = \begin{bmatrix} 2 & 4 & -2 \\ 1 & -1 & -3 \\ 3 & 12 & 1 \end{bmatrix}$.

6. $S = ($**siljish**$), H = ($**cho’zish**$), x =$**. HSx va**

$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & \end{bmatrix}$$

SHx ni hisoblang; har bir aylantirishni tavsiflang.

$$\begin{bmatrix} 2 & 4 \\ 2 & 3 \end{bmatrix}$$

[Manba: LA darsi, 3-hafta, masala 6]

Yechim. $Sx =$

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

, keyin HSx $=$

$$\begin{bmatrix} 2 & 4 \\ 2 & 3 \end{bmatrix}$$

; shuningdek $Hx =$

$$\begin{bmatrix} 2 & 4 \\ 2 & 3 \end{bmatrix}$$

, keyin SHx $=$

$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & \end{bmatrix}$$

. HS da siljish hissasi keyin cho'ziladi; $SHda$ esa avval cho'zish, keyin ustiga siljish qo'shiladi. Ikkala operatsiya bir xil, bosh tartibda, bosh natija: matritsalar almashmaydi.

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

,$v =$

$$\begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

: hisoblangu$^T v$ vauv^T; tekshiring $(uv^T)^T = vu^T$.

7. $u = 20 - 13$ **uchun**

[Manba: LA darsi, 3-hafta, masala 7]

Yechim. Ichki: $u^T v = 2 + 0 - 3 = -1$(skalyar). Tashqi: uv$^T =$

$$\begin{bmatrix} 2 & 0 & -1 \\ 0 & 0 & 0 \\ 3 & 0 & -3 \end{bmatrix}$$

(3×3). Element bo'yicha transponirlab, $(uv^T)^T = (vu^T)_{ij}$, shuning uchun $(uv^T)^T = vu^T$. Yozilgan matritsalar — $3 \times 3o$'zaro transponirlangan.

$$(AB)^T = B^T A^T$$

.

8. Masala $1danA, B$ **dan foydalanib,** $A^T B^T$ **mustaqil hisoblang; mos keladimi?**

$$[4]4 - 76$$

[Manba: LA darsi, 3-hafta, masala 8]

Yechim. $AB =$

$$\begin{bmatrix} 3 & 12 \\ -10 & -14 \end{bmatrix}$$

, demak $(AB)^T =$

$$\begin{bmatrix} 3 & -10 \\ 12 & -14 \end{bmatrix}$$

. Avval

$$\begin{bmatrix} 6 & 3 \\ -7 & 3 \end{bmatrix}\begin{bmatrix} 1 & 2 \\ -1 & 0 \end{bmatrix} = \begin{bmatrix} 3 & 12 \\ -10 & -14 \end{bmatrix}$$

transponirlangan: $A^T =$

$$\begin{bmatrix} -2 & 0 \\ 1 & 4 \end{bmatrix}$$

, $B^T =$

$$\begin{bmatrix} 4 & -7 \\ 6 & 3 \end{bmatrix}$$

; keyin $A^T B^T =$

$$\begin{bmatrix} -2 & 0 \\ 1 & 4 \end{bmatrix}\begin{bmatrix} 4 & -7 \\ 6 & 3 \end{bmatrix} = \begin{bmatrix} -8 & -14 \\ 28 & -5 \end{bmatrix}$$

. Hisoblaymiz: $(2\cdot 1 + (-1)(-2)) = 4, (2\cdot 3 + 0) = 6$

$$\begin{bmatrix} 4 & 6 \\ 28 & -5 \end{bmatrix}$$

$= (AB)^T$, bu (70) ni tasdiqlaydi. Lekin $A^T B^T =$

$$\begin{bmatrix} -8 & -14 \\ 28 & -5 \end{bmatrix}$$

farq qiladi: to’g’ri identifikator ko’paytuvchilar tartibini o’zgartiradi.

9. $E = \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}$ **deb belgilaymiz.** $\det E$ **ni hisoblang; uning ustunlariga tortilgan parallelogramm va uchburchkning maydonlarini toping; orientatsiya saqlanadimi?**

[Manba: LA darsi, 3-hafta, masala 9]

Yechim. $\det E = 3\cdot 2 - 1\cdot 1 = 5$. Parallelogramm maydoni $= |5| = 5$; uchburchk maydoni $= \frac{5}{2}$. Orientatsiya saqlanadi: $\det E > 0$.

10. $F = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 3 & 1 \\ 1 & 0 & 2 \end{bmatrix}$ **uchun** $\det F$ **ni birinchi qator kofaktorlari bo’yicha hisoblang; birinchi ustun bo’yicha tekshiring.**

$$\begin{bmatrix} 3 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 3 & 1 \end{bmatrix}$$

[Manba: LA darsi, 3-hafta, masala 10]

Yechim. Birinchi qator: $\det F = 2\det -1\det +0 = 2(6) - 1(-1) = 13.$ 0 2 1 2

$$\det F = 3\det\begin{bmatrix}0 & 2\\ 1 & 0\end{bmatrix} - 1\det\begin{bmatrix}1 & 2\\ 0 & 0\end{bmatrix} + 0\det\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix} = 3(0) - 1(0) + 0(1) = 0.(3.1)$$

Birinchi ustun: $2\det +1\det = 12 + 1 = 13$(kofaktor (3, 1) belgisi $0231(-1)^{3+1} = +$). Har qanday qator yoki ustun bir xil qiymat beradi.

11. 3×3 **matritsa** M **uchun** $\det M = 5$ **ma'lum.** N **ni hosil qiling: 1,2-qatorlarni almashtiring; 3-qatorni** $2ga$ **ko'paytiring;** $3 \times$ (**yangi 1-qator**)$ni2$**-qatorga qo'shing.** $\det N, \det(2M), \det(M^T M)$ **ni toping.**

[Manba: LA darsi, 3-hafta, masala 11]

Yechim. Almashtirish: belgi o'zgaradi (-5); 3-qatorni masshtablash: yana ko'paytuvchi 2; qator qo'shish: o'zgarishsiz. $\det N = -2 \cdot 5 = -10$. Barcha qatorlarni masshtablash: $\det(2M) = 2^3 \cdot 5 = 40$. Ko'paytuvchilar qoidasi: $\det(M^T M) = \det M^T \det M = 25$.

$$\det(M^T M) = \det M^T \det M = 25.$$

$= 0, v = 2, w = 1$. Hisoblang $(u \times v) \cdot w$, hajmlar

12. $u103$ **paralleliped va tetraedr hajmlari; agaru,**v **ni almashtirsak nima** o**'zgaradi?**

$$[2 \cdot 2 - 0 \cdot 14]$$

[Manba: LA darsi, 3-hafta, masala 12]

Yechim. $u \times v = [1 \cdot 1 - 2 \cdot 0;\ 2 \cdot 0 - 1 \cdot 3;\ 1 \cdot 3 - 2 \cdot 0] = \begin{bmatrix}1\\ -3\\ 3\end{bmatrix}$; w bilan skalyar ko'paytma: $1 \cdot 2 + (-3) \cdot 1 + 3 \cdot 4 = 2$ - $3 + 12 = 11$. Hajm

$$2 \cdot 2 - 0 \cdot 14$$

paralleliped 11; tetraedr $\frac{11}{6}$. $uvav$ ni almashtirish belgini -$11gao$'zgartiradi, lekin ikkala hajm $11va_6^{11}$(absolyut qiymatlar) bo'lib qoladi.

$$0 \cdot 0 - 1 \cdot 2 = -2$$

.

13. $u = \begin{bmatrix}1\\ 0\\ 1\end{bmatrix}$, $v = \begin{bmatrix}0\\ 2\\ 1\end{bmatrix}$, $w = \begin{bmatrix}1\\ 1\\ k\end{bmatrix}$ **uchun vektorlar koplanar bo'lgan barcha** k **ni toping; shu** $kdawniu, v$ **orqali ifodalang.**

$$0 \cdot 1 - 1 \cdot 2 - 2$$

$\Leftrightarrow\Rightarrow T = 0$**.** Bu **yerdau** $\times v = 1 \cdot 0 - 1 \cdot 1 = -1$**,**

[Manba: LA darsi, 3-hafta, masala 13]

Yechim. Komplanarlik

$$1 \cdot 2 - 0 \cdot 02$$

$T = -2 - 1+$ 2k = 2k $-3 = 0 \Rightarrow k = \frac{3}{2}$ degani. U holdaw $= \alpha u + \beta v$: birinchi ikki koordinata $\alpha = 1, 2\beta = 1 \Rightarrow \beta = \frac{1}{2}$ beradi; uchinchisini tekshiramiz: $1 + 1_2 = 3_2$ — mos keladi,$w = u + \frac{1}{2} v$.

$$W = U + (\frac{5}{7})V,$$

$$\begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} D = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} (16)$$

14. $D = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$, $X = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ **bo'lsin.** $DX = XD$ **bo'ladigan barcha** X **ni toping; agar** D **ning ikkala diagonal elementi teng bo'lsa nima bo'ladi?**

[Manba: LA darsi, 3-hafta, masala 14]

Yechim. $DX = \begin{bmatrix} 2a & 2b \\ 3c & 3d \end{bmatrix}$, $XD = \begin{bmatrix} 2a & 3b \\ 2c & 3d \end{bmatrix}$. Tenglik $b = c = 0$ talab qiladi, bunda $avad$ ixtiyoriy: aynan diagonal matritsalar bu D bilan kommutativ. Agar $D = \lambda I$ bo'lsa, u holda $DX = \lambda X = XD$ har qanday X uchun.

15. $u = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$, $v = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$, $w = \begin{bmatrix} 0 \\ 1 \\ 3 \end{bmatrix}$ $vaL = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ **bo'lsin. Qirralarni qayta hisoblamasdan,** $Lhar$ **bir qirraga qo'llanilgandan keyingi ishorali aralash ko'paytma va hajmni toping; hajm koeffitsienti nima uchun universal?**

$$\begin{bmatrix} 1 & 0 & 3 \\ 0 & 0 & -1 \end{bmatrix}$$

belgili aralash ko'paytma va hajmdan $Lnihar$ **bir qirraga qo'llaganidan keyin toping; nima uchun hajm koeffitsienti universal?**

[Manba: LA darsi, 3-hafta, masala 15]

Yechim. $U = \begin{bmatrix} u & v & w \end{bmatrix}$ deb olaylik; 12-masaladan, $\det U = 13$. O'zgargan qirralar LU ni hosil qiladi, shuning uchun $\det(LU) = \det L \det U = (-2)(13) = -26$; yangi hajm 26. $|\det L| = 2$ koeffitsienti U dan mustaqil, chunki $\det(LU) = \det L \det U$ har qanday U uchun — determinantlar ko'paytmasi qoidasi geometrik ma'noni anglatadi: $|\det L|$ — hajmni o'zgartirish koeffitsienti.

5 4-hafta — Matritsa rangi va teskari matritsa (Linealalgebra I)

Lineal algebra I kursining 4-haftasi matritsa rangi va Gauss-Jordan usuli bilan matritsani teskarisini olishni $o'z$ ichiga oladi, 1-3 haftalar bo'yicha Test 1 bilan tugaydi. Repozitoriyda 4-hafta uchun alohida mashqlar varaqasi yo'q, shuning uchun bu qism (i) Strangning berilgan materialidan to'plangan ikki mavzuga to'liq formulalar katalogi va $(ii)TA$ kursi varaqalari uslubida gradatsiyalashgan mashq to'plamini to'liq yechimlari bilan beradi. Ular kursning boshqa joylarida bir xil matematik tuzilmalardan foydalanadi, shuning uchun bir vaqtda 1-testga matritsa-algebra qismlari bo'yicha tayyorgarlik xizmatini qiladi.

5.1 4-hafta formulalar katalogi

6.1.1 Rang va chiqarib tashlash

rank$A \le \min(m,n), (106) A \in R^{n\times n}$: rank$A = n \Longleftrightarrow Ax = 0$ faqatx $= 0$ ni beradi, $(107) A$ teskari $\Longleftrightarrow$ rank$A = n \Longleftrightarrow n$ ta ustun elementi $\Longleftrightarrow$ det $\not A = 0, (108) A$ teskari $\Longleftrightarrow Ax = b$ har qandayb uchun ayni bitta yechimga ega. (109)

6.1.2 Teskari matritsa (Strang, Introduction to Linear Algebra 4e, §2.5)

$AA^{-1} = I = A^{-1}A$; $x = A^{-1}bAx = b$ ni yechadi, $(110) A^{-1}$ mavjud $\Longleftrightarrow$ chiqarib tashlash n ta ustun elementini beradi, (111)

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = (1)/(ad-bc)\begin{bmatrix} d & b \\ c & a \end{bmatrix}(ad-bc \neq 0),$$
$$(112)$$

(teskari tartib), $(113) diag(d1,\ldots,$ **dn**$) - 1 = diag(1/d1,\ldots,1/dn)$**.**

$$diag(d_1,...,d)^{-1} = diag(1/d_1,...,1/d)(d \neq 0), (114)$$

$\not x = 0, Ax = 0 \Longrightarrow A$ teskari emas, $(115) BA = I, AC = I \Longrightarrow B = B(AC) = (BA)C = C$(yagonalik). (116) Talqin (115): hech qanday matritsa$0nix$ ga qaytara olmaydi.

6.1.3 Gauss-Jordan chiqarib tashlash

$$elementarqatorlar_{-1} - - - - - - - - - - - - - - - - - - \to \begin{bmatrix} I & A \end{bmatrix}$$

$\begin{bmatrix} A & I \end{bmatrix}$ (117) Bir xil qator operatsiyalari $AniIgavaIniA^{-1}$ ga aylantiradi. Qator operatsiyalari: ikki qatorni almashtirish; qatorni nol bo'lmagan konstantaga ko'paytirish; boshqa qatordan ko'paytmani qo'shish. Har biriga oddiy teskarisiga ega bo'lgan elementar matritsa E to'g'ri keladi, shuning uchun $(E_2E_1A)^{-1} = A^{-1}E_1^{-1}E^{-1}$ teskari tartibda chiqadi. 2

5.2 4-hafta mashqlari (rang, teskari, Gauss-Jordan) — savollar vato'liq yechimlar

1. $A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 5 \\ 3 & 6 & 4 \end{bmatrix}$ **uchun chiqarib tashlash orqali** rank$A$ **ni hisoblang.**

[Manba: $LA4$-hafta amaliyoti, masala 1]

Yechim. Birinchi ustun elementi (1) ostida chiqaramiz: $R_2 - 2R_1 \to (0,0,3)$; $R_3 - 3R_1 \to (0,0,1)$. Ikkinchi ustun elementlari yo'qoldi, demak ikkinchi ustun elementi 3-ustundan keladi: ustun elementi $= 3$. Uning ostida chiqaramiz: $R_3 - \frac{1}{3} R_2 \to (0,0,0)$. Ikkita nol bo'lmagan ustun element: rank$A = 2$. E'tibor bering $col_2 = 2col_1$, bu boshidan ko'rinib turardi — rang kamchiligi belgisi. 2 1

2. $B =$**, uchta usul bilan teskari ekanligini aniqlang: ustun elementlar, 4 2 determinant va nol bo'lmani nol vektor.**

[Manba: LAming 4-haftalik amaliyoti, 2-masala]

Yechim. (Asosiy)$R_2 - 2R_1 \to (0,0)$: faqat bitta asosiy — noyob. (Determinant)$ad - bc = 2\cdot 2 - 1\cdot 4 = 0$ — (112) bo‘yicha noyob. $(Nol-$vektor$)B(1,-2)^T = (2-2, 4-4)^T = 0$ — $Bx = 0$ tenglamaning nol bo‘lmagan yechimi, demak (115) bo‘yicha B teskarisiz. Uchala test ham mos keladi. 212×2 formulasi bo‘yicha hisoblang va $AA^{-1} = I$ ekanligini tekshiring.

3. $A = \begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}$ **matritsasining teskarisini toping.**

[Manba: LAming 4-haftalik amaliyoti, 3-masala]

Yechim. $\det A = 6 - 5 = 1$**, shuning uchun (112) bo‘yicha** $A - 1 = \begin{bmatrix} 3 & -1 \\ -5 & 2 \end{bmatrix}$**.**
Tekshirish: $A \cdot A - 1 =$ **[**$2 \cdot 3 + 1 \cdot (-5), 2 \cdot (-1) + 1 \cdot 2$**;** $5 \cdot 3 + 3 \cdot (-5), 5 \cdot (-1) + 3 \cdot 2$**]** $= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$**.**

$$A^{-1} = (\frac{1}{7}) \cdot \begin{bmatrix} 2 & -3 \\ 1 & 5 \end{bmatrix}$$

$$A = [5 \cdot 3 + 3 \cdot (-5) 5 \cdot (-1) + 3 \cdot 2; 01] = \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}$$

4. $\begin{bmatrix} A & I \end{bmatrix}$ **matritsiga Gaus-Jordan usulini qo‘llab** $A =$**. 3 7 matritsasini teskarisini toping.**

[Manba: LAming 4-haftalik amaliyoti, 4-masala]

Yechim.

$$\begin{bmatrix} 1 & 2 \\ 3 & 7 \end{bmatrix} \to R_2 - 3R_1 \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \to R_1 - 2R_2 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix},$$

$$A^{-1} = \begin{bmatrix} -7 & 2 \\ 3 & -1 \end{bmatrix}$$

Shunday qilib $A^{-1} =$; tekshirish: $AA^{-1} ==$.

1

5. Gaus-Jordan usuli bilan teskarisini toping va tekshiring, 0 −1 2 asosiy ko‘paytmalari $\det K$ **ga teng ekanligini.**

[Manba: LAming 4-haftalik amaliyoti, 5-masala (Strangning sevimli misoli)]

Yechim. [$K|I$**] da ishlaymiz: birinchi opor 2 bilan chiqaramiz** $(R2 + (\frac{1}{2})R1)$**, so’ngra ikkinchi opor** $\frac{3}{2}(R3 + (\frac{2}{3})R2)$**, so’ngra uchinchi 4/3; yuqoriga davom etib, oporlar ustidini tozalaganda** o**’ngda** $K - 1$ **hosil bo’ladi.**

$$K^{-1} = (\frac{1}{4}) \cdot \begin{bmatrix} 3 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 3 \end{bmatrix}$$

bo‘ladi. Asosiy ko‘paytmalar $2 \cdot \frac{3}{2} \cdot 4_3 = 4 = \det K$ ga teng, va haqiqatan ham K^{-1} butun sonli matritsani $\det K ga$ bo‘ladi — (112) dan keyin belgilangan namuna. (K simmetrik va uch diagonal, lekin K^{-1} zich.)

4-masaladagi A^{-1} yordamida.

6. $Ax = b$ **tenglamani** $A =, b = 371$

uchun yeching.

[Manba: LAming 4-haftalik amaliyoti, 6-masala]

Yechim. $x = A-$**1b** $= \begin{bmatrix} 7 & -2 \\ -3 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix} =$ **[**$7 \cdot 1 - 2 \cdot 1$**;** $-3 \cdot 1 + 1 \cdot 1$**]** $= \begin{bmatrix} 5 \\ -2 \end{bmatrix}$**.**

$$A^{-1} = (1)/(7) \begin{bmatrix} 1 & -2 \\ 3 & 1 \end{bmatrix}$$

; $(AB)^{-1}$ ni avval

7. $A = vaB = 0211$ **matritsalar berilgan.** $(AB)^{-1}$ **ni avval**

[Manba: LAming 4-haftalik amaliyoti, 7-masala]

Yechim. (113) formulaga ko’ra, $(AB)^{-1} = B^{-1}A^{-1}$. Bu yerda $\det A = 2va \det B = 2$, shuning uchun

$$det(AB) = 4, teskarimatritsani (AB)^{-1} = (1)/(4) \begin{bmatrix} 2 & -3 \\ -2 & 4 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} [(-121)/2(21)/(-2);$$

$$(-314)/4(21)/(-2)] = [\frac{1}{2}\frac{1}{2}; \frac{3}{4}\frac{1}{4}] — birxil matritsa. Va$$

$(AB) - 1(AB) =$ [1/2 1/2; 3/4 1/4] $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ = [1/2 1/2; 3/4 1/4] * [(−1 2 1)/2 (2 1)/(−2); (−3 1 4)/4 (2 1)/(−2)] = $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$. Teskari tartib qonuni tasdiqlandi.

8. (Tushunchaviy, Test-1 uslubida.)$To’g’ri$ **yoki noto’**$g’ri$**, asoslari bilan:** (a) **Agar** A **matritsa teskari bo’lsa,** A^T **ham teskari bo’ladi.** (b) **Agar** $Ax = \lambda xx \neq 0va\lambda = 0$ **bo’lsa,** A **teskari emas.** (c) **Teskari** 3×3 **matritsalar ko’paytmasi noyob bo’lishi mumkin.** (d) **Har qanday** 2×3 **matritsa uchun** $\text{rank}(A) \leq 2$**.**

[Manba: LAming 4-haftalik amaliyoti, 8-masala]

Yechim. $(a)To’g’$ri: $Adan$ ta asosli element bo’lsa $\Leftarrow\Rightarrow A^T dan$ ta asosli element bo’ladi (qator rangi = ustun rangi); rasmiy ravishda $(A^T)^{-1} = (A^{-1})^T$. $(b)To’g’$ri: $Ax = 0 \cdot x = 0$ tenglamani qanoatlantiruvchi nol bo’lmagan $x(115)$ shartni buzadi. (c) Noto’$g’$ri: (113) formulaga ko’ra teskari matritsalar ko’paytmasi ham teskari bo’ladi. $(d)To’g’$ri: rang matritsa satrlari sonidan oshmaydi ($harbir$ asosli element bitta satrni egallaydi), shuning uchun rank $\leq 2 < 3$.

9. (Amaliy, Test-1 uslubida.)c **ning qanday qiymat**$(lar)ida A = \begin{bmatrix} 1 & c \\ 4 & 2 \end{bmatrix}$ **matritsasi yagona (singulyar emas)? Qolgan qiymatlar uchun** $Ax = \begin{bmatrix} 6 \\ 4 \end{bmatrix}$ **ni yeching.**

[Manba: LAming 4-haftalik amaliyoti, 9-masala]

Yechim. $\det A = c - 6$; noyob bo'lishi uchun aynan $c = 6$. Aks holda $A^{-1} =^1$

$$\det A = (c-6)/(4c-20)$$

vax $= A^{-1}b =^1$, masalan $c = 7$ da:$x = (8, -2)^T$.

$$x = A^{-1}b = (beginarrayc8 - 2endarray).$$

10. (Yuqori murakkablik.) Isbotlang, agar A teskari bo'lsa, u holda $\operatorname{rank}(AB) = \operatorname{rank}(B)$ har qanday mos keladigan B uchun.

[Manba: LAm 4-hafta amaliyoti, 10-masala]

Yechim. Teskari matritsaga ko'paytirish soni asosli vektorlarni o'zgartirmaydi: standavy fikrlash: $\operatorname{rank}(AB) \leq \operatorname{rank}B(AB$ qatorlari — B qatorlarining chiziqli kombinatsiyalari), va xuddi shuni $B = A^{-1}(AB)$ uchun qo'llaymiz, $\operatorname{rank}B \leq \operatorname{rank}(AB)$ hosil bo'ladi; ikkala tengsizlik tenglikni talab qiladi. Intuitsiya: teskari ko'paytuvchi faqat fazani cho'zadi/aylantiradi/siljitadi, lekin tekislamaydi, shuning uchun rang yo'qolishi yoki paydo bo'lishi mumkin emas.

6 Darsliklardan qo'shimchalar — to'liq yechimlari bilan-majburiy o'qish uchun qo'shimcha mashqlar

6.1 Strang, Introduction to Linear Algebra 4e — tanlanganmasalalar yechilgan

1. (§1.1) Barcha cu ni tavsiflang $u = (2, 1)$ uchun va barcha cu + dv kombinatsiyalarini $u = (2, 1), v = (1, 2)$ uchun; qachon cu + dv kombinatsiyalari tekislikni to'ldiradi?

[Manba: Strang Intro 4e, g'oyalar to'plami §1.1]

Yechim. cu (2, 1) orqali o'tgan to'g'ri chiziqni to'ldiradi: undagi har bir vektor — uning biror karrali. cu + dv koordinata boshidan o'tgan tekislikni to'ldiradi, agar $21u$ vav kollinear bo'lmasa; bu yerda det $= 3\not= 0$, shuning uchun ikki yo'nalat 1 2 mustaqil va kombinatsiyalar butun R^2ni to'ldiradi. Aynan shu mezon — nol bo'lmaydigan determinant — 4-haftada teskarilik sifatida qaytadi (108).

2. (§1.2) Skalyar ko'paytmalarni hisoblang:$u \cdot v$ uchunu $= (1, 2, 1), v = (3, 0, 2)$; ikkalasiga perpendikulyarw vektorni toping.

[Manba: Strang Intro 4e, masalar to'plami §1.2]

Yechim. $u \cdot v = 3 + 0 + 2 = 5$. Perpendikulyar vektor — vektor ko'paytmasi:$w = u \times v =$

$$\begin{bmatrix}1\\2\\1\end{bmatrix} \times \begin{bmatrix}3\\0\\2\end{bmatrix} = [(2 \cdot 2 - 1 \cdot 0);$$

$$(1 \cdot 3 - 1 \cdot 2); (1 \cdot 0 - 2 \cdot 3)] = \begin{bmatrix}4\\1\\-6\end{bmatrix}$$

; tekshirish: $w \cdot u = 4 + 2 - 6 = 0 va w \cdot v = 12 + 0 - 12 = 0$**.**

3. (§1.2) Qachonu $\cdot v = v \cdot u$ **manfiy va bu burchak haqida nima deydi? Nolga teng qachon?**

[Manba: Strang Intro 4e, tushuncha §1.2]

Yechim. Manfiy aynan $\cos\theta < 0$ bo'lganda, ya'ni burchak 90° dan katta (o'tkir); nol $\theta = 90^\circ$ da — perpendikulyarlik. Bu tasnif, 2-haftadagi TA masalarida qo'llanilgan. $21, b = (3,4)^T$ orqali A^{-1}.

4. $(\S 2.5) Ax = b$ **ni yeching** $A = 12$

$$[2 - 12; 3 - 42; \frac{2}{3}]$$

[Manba: Strang Intro 4e, masalar to'plami §2.5]

Yechim. $\det A = 3$: $A^{-1} = ^1; x = A^{-1}b = ^1 = ^1 =$.

$$detA = 3; A = [1 - 12; 3 - 38; 35\frac{5}{3}];$$
$$x = A^{-1}b = [\frac{1}{4}; 4; 2] = [\frac{14}{4}; 4; 2] = [\frac{7}{2}; 4; 2](16)$$

Tekshirish: $2 \cdot (\frac{2}{3}) + 5 \cdot (\frac{1}{3}) = 3, 2 \cdot (\frac{1}{3}) + 2 \cdot (\frac{5}{3}) = 4$**.**

5. (§2.5) Nol bo'lmagan $Ax = 0$ **yechimi teskarini istisno qiladi? Ikkita jumladan iborat fikrni keltiring, konsept «hech qanday matritsa**$0nix$ **ga qaytara olmaydi» deb nomlanadi.**

[Manba: Strang Intro 4e, §$2.5ga4$-e'lon]

Yechim. Agar $Ax = 0x \not= 0 va A$ da teskari bo'lsa, u holdax $= A^{-1}Ax = A^{-1}0 = 0$ — ziddiyat. Nol vektorning «teskari tasviri» faqat aynilikda mavjud; nol-vektor buni emasligini isbotlaydi.

6. (§2.5) Agar $AvaC$ **teskari** 3×3 **matritsalar bo'lsa,** $A + C$ **teskari bo'ladimi?** AC**?** $A^{-1}C$**?**

[Manba: Strang Intro 4e, masalar to'plami §2.5]

Yechim. $A + C$: albatta emas (masalan, $C = -A0$ beradi); teskari matritsalarning yig'indisi aynan bo'lishi mumkin. AC: ha, $(AC)^{-1} = C^{-1}A^{-1}(113)$. $A^{-1}C$: ha, ikkita teskari matritsaning ko'paytmasi.

6.2 Strang, Linear Algebra and its Applications 4e, §1.6 (Teskari va-transponirlangan) — asosiy natijalar va tahlil qilingan misol

Belgilangan o'qish §1.6 katalogga uchta qo'shimcha element qo'shadi: Gauss– 2 1 1

$$\det A^{-1} = (1)/(\det A)$$

Jordani bilan $A = 4 - 60_{detA}$ va transponirlanish -2 7 2 identifikatorlari. 2 1 1

1. $A = \begin{bmatrix} 2 & 1 & 1 \\ 4 & -6 & 0 \\ -2 & 7 & 2 \end{bmatrix}$ **uchun Gauss-Jordan usulini bajaring va** $A - 1ni$ **ko'rsating.**

[Manba: Strang LAiA 4e, §1.6, misol 1]

Yechim. Ini to'ldirib, chiqarib tashlaymiz: asosiy $2, -8, 1 (R_2 - 2R_1, R_3 + R_1$, keyin $R_3 + \frac{3}{4}R_2$ amallaridan so'ng). Yuqoriga davom etib, asosiy elementlar ustini tozalab, satrlarni $2, -8, 1 ga$ bo'lib,

$$A^{-1} = (\frac{1}{8}) \begin{bmatrix} 12 & -5 & -6 \\ 8 & -3 & -2 \\ -1 & 1 & 1 \end{bmatrix}$$

hosil qilamiz. Determinant asosiy elementlar ko'paytmasiga teng $2(-8)(1) = -16$, va haqiqatan ham A^{-1} det Aga bo'lishni ko'rsatadi (kofaktorlarning belgilari hisobga olingan — $A^{-1} = \frac{adjA}{detA}$ ga ekvivalent).

2. §$1.6da$ qo'llanilgan determinant-transponirliqlik identifikatsiyalarini ko'rsating: $\det A^T, \det(AB), \det A^{-1}$**.**

[Manba: Strang LAiA 4e, §1.6]

Yechim. $\det A^T = \det A$; $\det(AB) = \det A \det B (1-$jadval); va $B = A^{-1}$ deb olib, $\det(A) \det(A^{-1}) = \det I = 1$, shuning uchun $1 \det A^{-1} = \det A$, bu esa "teskari mavjud $\Leftrightarrow\Rightarrow \det A \neq 0$" degani aniq, determinantlar katalogini (3-hafta) teskari matritsalar katalogi (4-hafta) bilan bog'laydi.

3. Isbotlang $(A^{-1})^T (A^T)^{-1} va A^T A$ **simmetrik ekanini tushuntiring, manfiy bo'lmagan diagonalga ega.**

[Manba: Strang LAiA 4e, §1.6]

Yechim. $AA^{-1} = I$ ni transponirib, $(A^{-1})^T A^T = I$ hosil bo'ladi, bu esa

$$(A^{-1})^T = (A^T)^{-1} (16)$$

degani aniq: Simmetriya: $= A^T A$; diagonal elementlari P

$$(A^T)^{-1} = (A^{-1})^T; matritsa A^T A simmetrik,$$
$$(A^T A)^T = A^T A, uning diagonal elementlari manfiy$$

$$bo'lmagan : a_{ii} = \Sigma_{k=1}^{n} a_{ki}^2 \geq 0 — kvadratlar yig'indisi — fakt. (17)$$

3-hafta ma'ruzalaridagi CC^T bilan bog'liq.